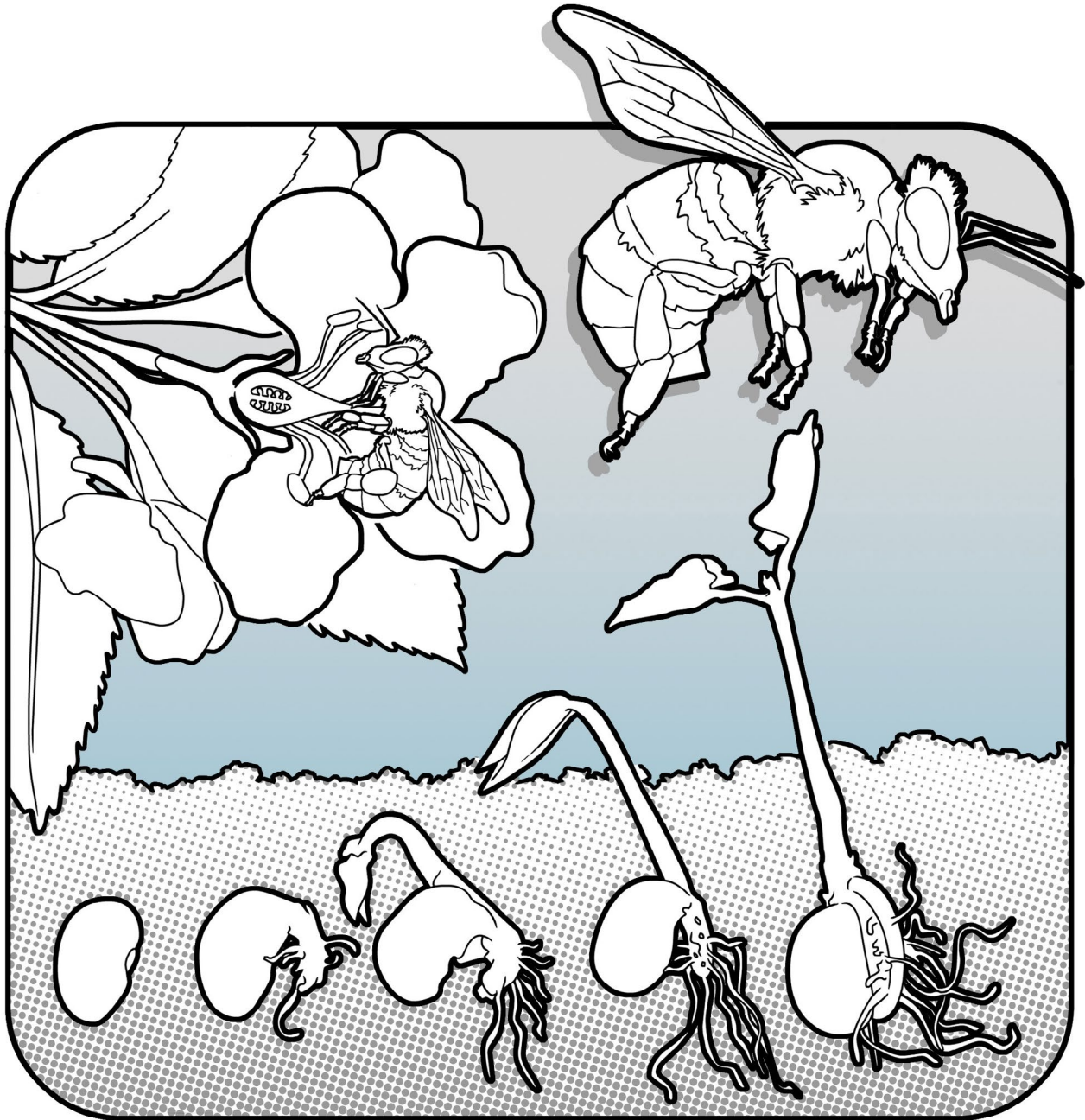


GRADE 8

NATURAL SCIENCES

WORKBOOK



MODULE 1: LIFE AND LIVING



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LIFE AND LIVING

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GRAPHS

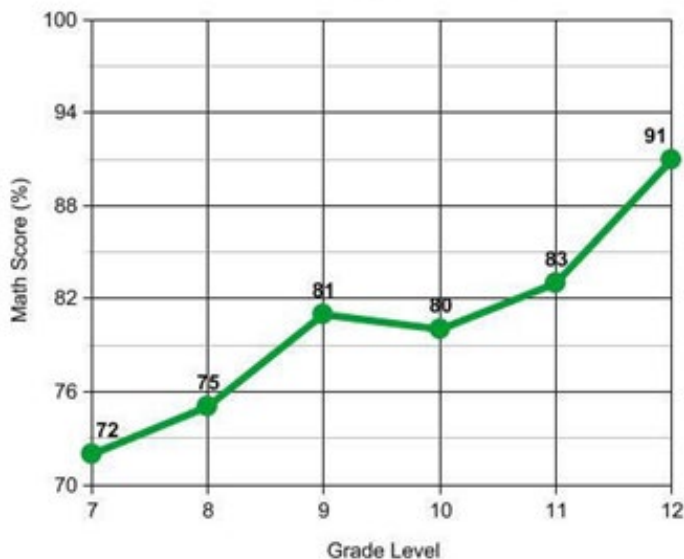
Introduction

We are going to learn how to analyse and draw four different types of graphs: line graphs, bar graphs, histograms and pie graphs. Graphs are an important tool in maths, science and biology to communicate information.

Analysing graphs

Line graph 1

Line graph of Jill's maths scores from grade 7-12

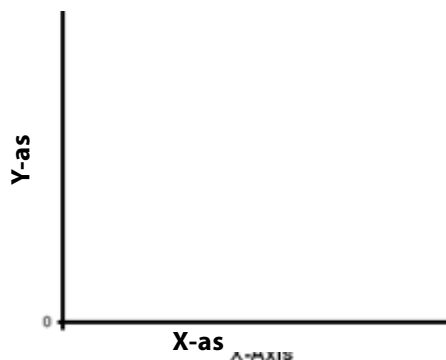


What does this graph show us? You may be asked to DESCRIBE the graph or to EXPLAIN the graph.

When you are asked to describe the graph, you need to clarify what is happening on the graph.

So, describe the graph above:

In some instances, it will also be important to EXPLAIN the graph. When you are asked to explain the graph, you need to say WHY it is happening. This is not really relevant to the graph above, but we will look at another example below where it is relevant.



In every graph that is drawn with an x-axis and a y-axis there is an independent variable and a dependent variable.

The x-axis is horizontal and depicts the independent variable. The y-axis is vertical and depicts the dependent variable.

The independent variable is the variable that the graph is showing that does not depend on anything else. The dependent variable is the variable that the graph is showing that depends on the independent variable.

In the example above, what is the independent variable and what is the dependent variable?

The maths score **DEPENDS** on the grade, therefore it is the dependent variable. The grade that the learners are in **DOES NOT DEPEND** on the maths score, therefore it is the independent variable.

In a graph that shows plant height over time in days, what is the independent variable and what is the dependent variable?

This is because time does not depend on the plants growing, and plant height will change with time, so it depends on time.

In a graph that shows how the density of water changes with temperature, what is the independent variable and what is the dependent variable?

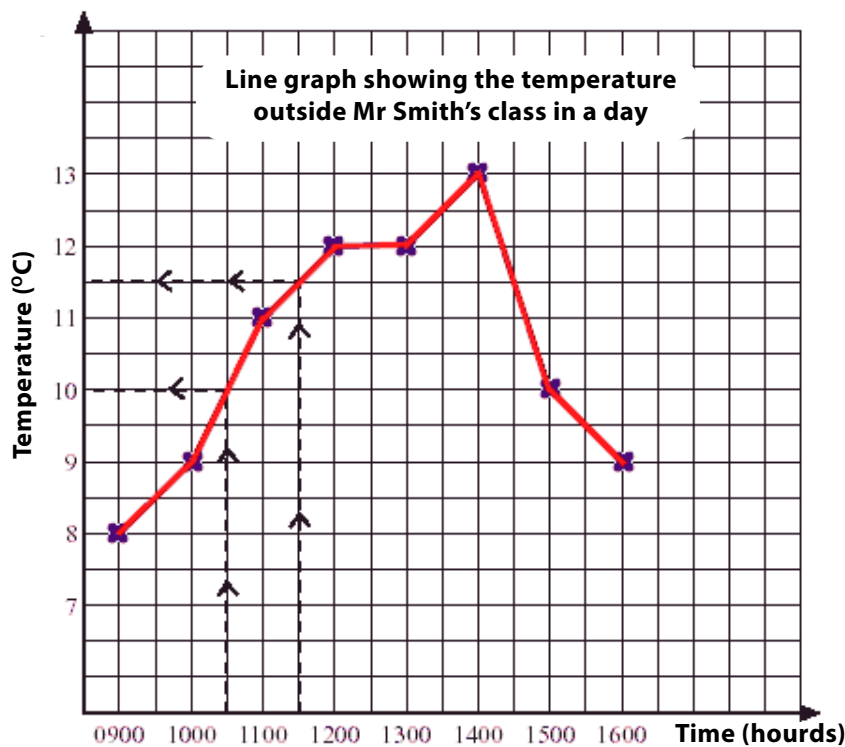
This is because temperature does not depend on the density of the water, and the water density will change with temperature, so it depends on temperature.

Line graph 2

Mr Smith recorded the temperature outside his classroom every hour during one school day. His results are listed in the following table:

Time	0900	1000	1100	1200	1300	1400	1500	1600
Temperature ($^{\circ}\text{C}$)	8	9	11	12	12	13	10	9

He decided to plot a line graph to represent the data. Line graphs are useful for showing how things change over time.



Describe the graph:

What is the independent variable and what is the dependent variable? Explain.

Based on the graph:

What is the temperature outside the classroom at 10:30? _____

What is the temperature at 11:30? _____

What is the temperature at 13:00? _____

What is the temperature at 13:30? _____

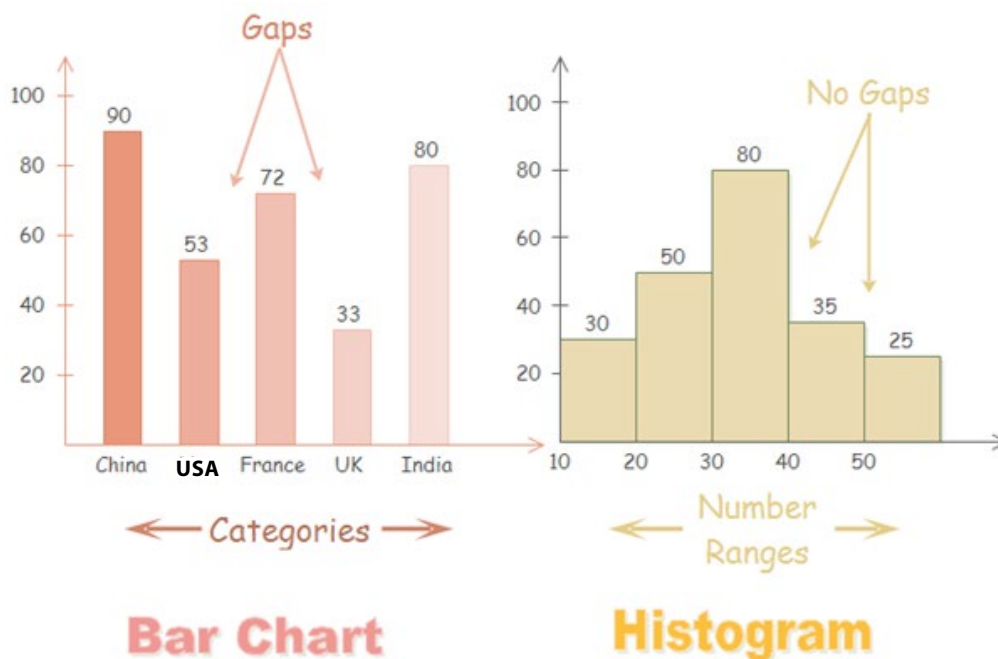
When the temperature is 8 °C, what time is it? _____

When the temperature is 10 °C, what time is it? _____

Now let's try to explain the graph:

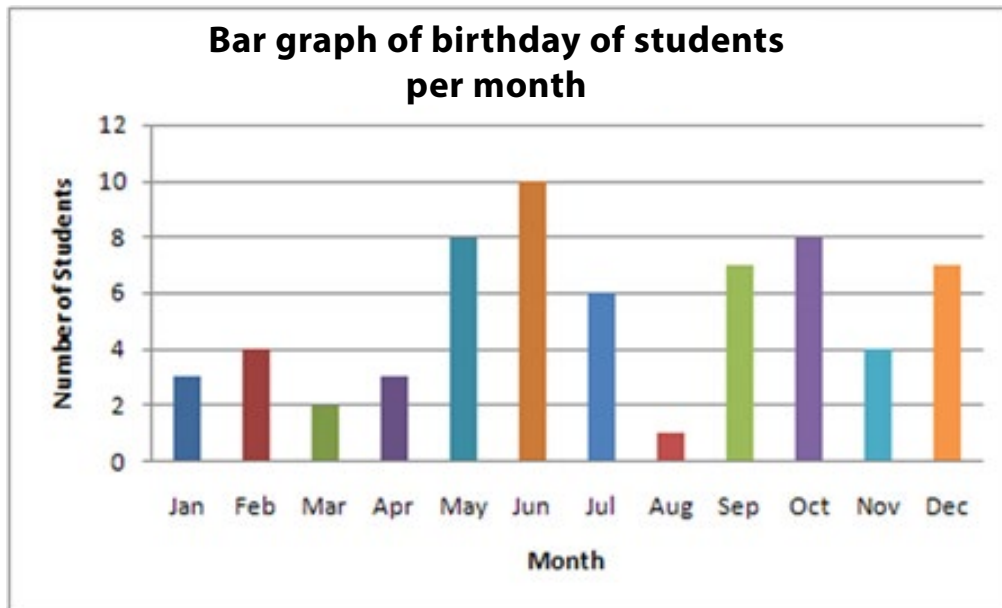
Bar graphs (charts) and histograms

Bar graphs and histograms are useful for grouping information. The difference between a bar graph and a histogram is that a bar graph represents categories and there are gaps between the bars. A histogram shows a continuous range of numbers and there are no gaps between the bars. This is shown below :



In a bar graph, we put the categories/groups on the x-axis. The categories/groups are always the independent variable.

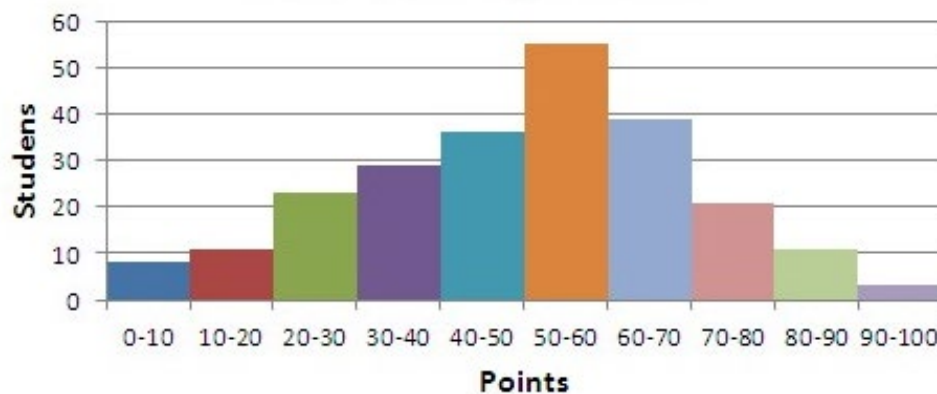
Look at the bar graph below:



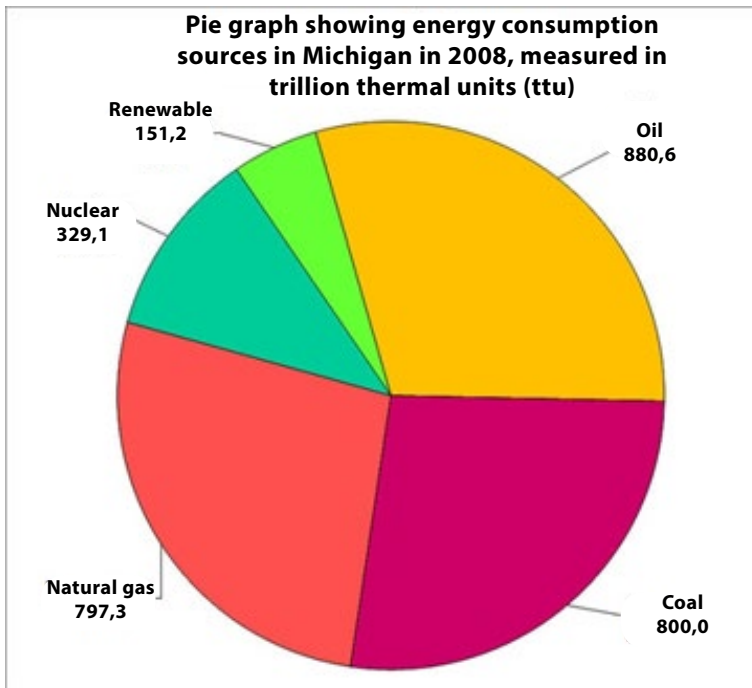
Describe the graph:

Look at the histogram below:

Histogram of points scored on a test by a group of students



Describe the graph:



Pie graph

A pie graph (also known as a pie chart) also groups information, like a bar graph or a histogram, but it is represented like a pie with slices.

Describe the graph:

Note that every graph has a **title**. You need to mention what type of graph it is and explain in brief what the graph is showing. Go back to the graphs above and note the titles.

Assignment 1: Graph exercises

Complete the three graph exercises and answer the questions. The line graph and bar graph should be completed on graph paper.

Line graph

In the pancreas, there are glands that produce insulin. Insulin is a hormone that gets released when there is a high level of glucose (a type of sugar) in the blood. This happens after a meal. Insulin will stop being released once the blood sugar level is normal.

The pancreas of a person with diabetes does not produce insulin, or it only produces very little insulin. If there is not enough insulin being produced by the glands of the pancreas, the amount of glucose in the blood will remain high.

A blood glucose level above 140 mg/dL for an extended period is not considered normal. This disease, if not brought under control, can lead to severe complications and even death.

1. Draw a line graph of blood glucose level over time in persons A and B based on the data below. Use the graph paper provided. You need to plot two sets of data separately on the same graph. **(16)**

Time after eating (hours)	Glucose of blood in Person A (mg/dL)	Glucose of blood in Person B (mg/dL)
0,5	170	180
1	155	195
1,5	140	230
2	135	245
2,5	140	235
3	135	225
4	130	200

(mg is a milligram, which is a thousandth of a gram; a dL is a decilitre, which is a tenth of a litre)

Assessment grid for line graph

Title	2 marks
X-axis label	2 marks
Y-axis label	2 marks
X-axis scale	1 mark
Y-axis scale	1 mark
Check 4 points (2 on each graph) to see that they are plotted correctly	4 marks
Key	2 marks
Neatness	1 mark
Size	1 mark

2. What is the dependent variable? Explain. **(2)**

3. What is the independent variable? Explain. **(2)**

4. Which, if any, of the above individuals (A or B) has diabetes? Explain. (2)

Bar graph

In the Kruger National Park, the elephant population is unstable. This is due to culling or removing of the elephants from the park to other ecosystems when their numbers become too large. Elephant culling was stopped in 1994.

5. Draw a bar graph of the elephant population size in the Kruger National Park over time based on the data below. Use the graph paper provided. (15)

Year	Elephant population size
1947	560
1960	1 250
1967	6 500
1985	8 200
2004	11 500
2010	13 000

Assessment grid for bar graph

Title	2 marks
X-axis label	2 marks
Y-axis label	2 marks
X-axis scale	1 mark
Y-axis scale	1 mark
Check that the 6 bars are plotted correctly	3 marks
Check that the 6 bars are labelled correctly	2 marks
Neatness	1 mark
Size	1 mark

6. What is the dependent variable? Explain. (2)

7. What is the independent variable? Explain. (2)

Pie graph

In 2012, 70% of South Africa's total primary energy consumption came from coal, followed by oil (22%), natural gas (4%), nuclear (3%) and renewables (1%, primarily from hydropower), according to the BP Statistical Review of Energy 2013. South Africa's dependence on coal has led the country to become the leading carbon dioxide emitter in Africa and the 14th largest in the world, according to the latest (2011) estimates by the American Energy Information Administration (source: www.eia.gov).

8. Draw a pie graph based on the information provided above. Remember to provide a relevant title and to show all your calculations. (9)

Instructions for drawing a pie graph

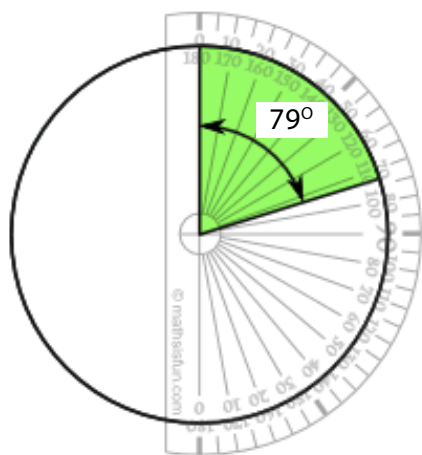
You need to draw the pie graph using a compass to get a perfect circle. Now you need to get the correct division.

The total percentage is 100%. For coal your calculation will be:

$70/100 \times 360^\circ = 252^\circ$ – if your answer is not a whole number, round off the number to the nearest whole number. This number is the angle that you will divide your pie graph into to represent the coal usage.

You use the same calculation for all the other energy types, e.g. $22/100 \times 360^\circ = 79^\circ$.

You now need to divide your pie graph using a protractor, as shown below:



You need to make sure that the person reading the graph can see which section represents which energy type. You may use labels inside the graph, or a key.

The pie graph should never indicate the degrees that you worked out. The labels or the key should show the

real values (e.g. 70%, **NOT** 252°).

Assessment grid for pie graph

Title	2 marks
Pie graph was drawn with a protractor	1 mark
Correct division of the pie graph	2 marks
Key or label of the portions of the pie graph	2 marks
Neatness	1 mark
Size	1 mark

Total = 50 marks

PHOTOSYNTHESIS AND RESPIRATION

Introduction

Living organisms need energy to sustain life. We are going to learn where this energy comes from and how it is made available to living things.

Photosynthesis

Living organisms interact with one another and are interdependent. What does this mean?

What is the most important way that organisms interact with one another?

For example, the buck eats the grass and the lion eats the buck. The interactions and interdependence between organisms in an ecosystem is driven by the need for energy to sustain life. We are going to learn about where this energy comes from.

The Sun is the ultimate source of energy

The Sun is the ultimate source of energy for all living things. The Sun provides light energy. _____ energy is also known as _____ energy.

All living organisms are dependent on energy in order to live. However, the light energy from the Sun cannot be used by living organisms directly. It needs to be converted into a form that can be used by living organisms.

What is the basic principle of energy?

During the process of photosynthesis, green plants convert _____ energy from the Sun into _____ energy, which living organisms can use.

The process of photosynthesis

Photo means _____. Synthesis means _____. So, during the process of photosynthesis, plants use the Sun's light to make their own food in a series of chemical reactions.

Plants require the following resources in order to photosynthesise:

- Sunlight – plants are green in colour because they have a green pigment inside their cells. This pigment is called _____. Chlorophyll absorbs the radiant energy from the Sun.
- Water – the _____ of the plant absorb water from the _____.
- Carbon dioxide – plants absorb carbon dioxide from the _____, which is absorbed through plant leaves.

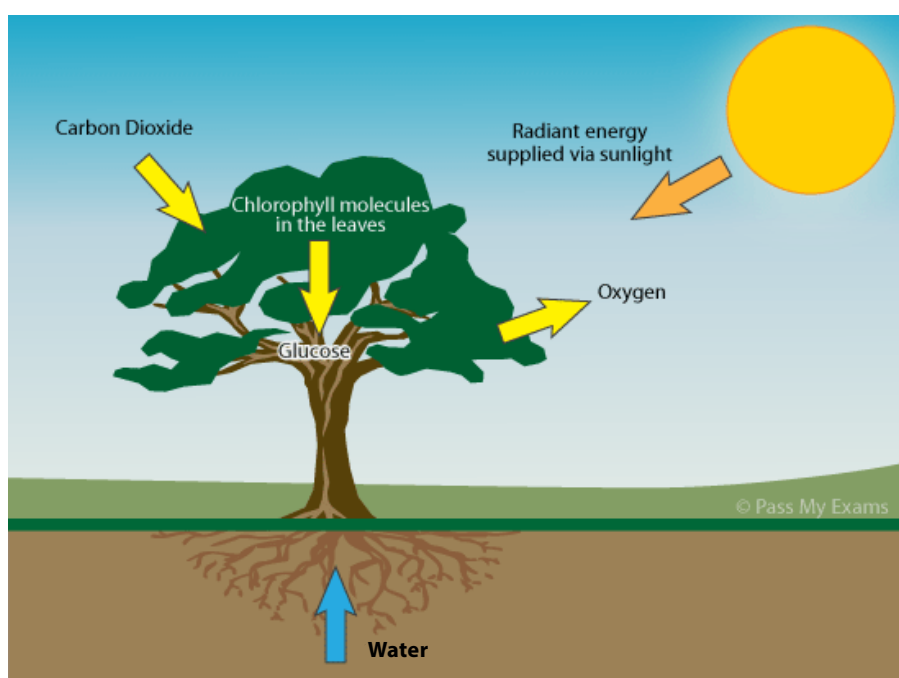
In the process of photosynthesis, _____ is made by the plant. Oxygen

is also produced in the process of photosynthesis. What happens to the oxygen?

The chemical reaction for photosynthesis is:

Chlorophyll

Carbon dioxide + water
+ light _____
→ Glucose + oxygen



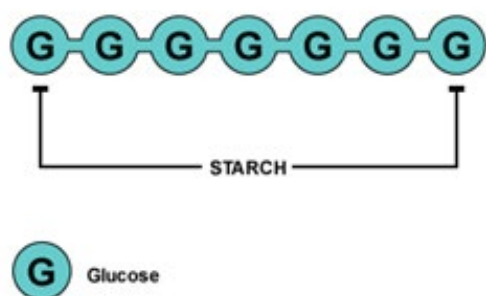
The process of photosynthesis

The _____ energy in the Sun is converted into _____ energy. The chemical energy that is stored in the plant is _____ energy. When animals eat the plants, they consume the potential chemical energy that the plants produced during photosynthesis.

Plants convert glucose into other compounds

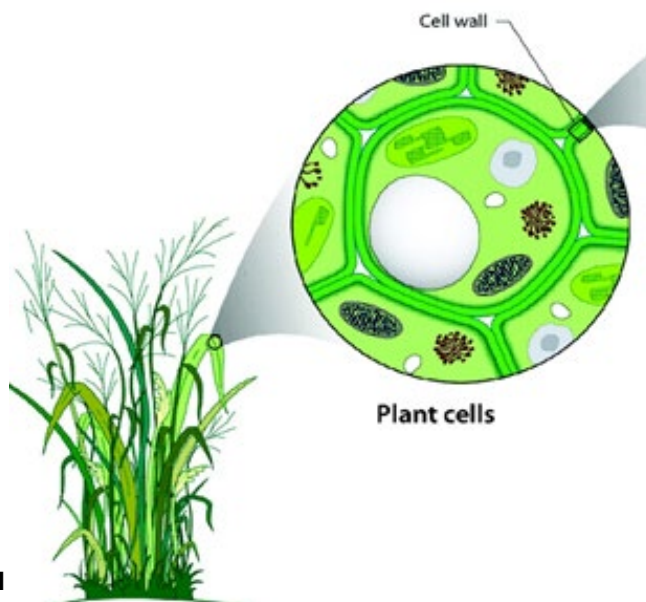
Glucose is a simple sugar that is made by plants during the process of photosynthesis. Glucose is used by plants for food. Food gives plants energy. This energy is used for life processes such as _____.

Not all the glucose is used by the plant in its life processes. Some of the glucose is stored in the plant as _____. Starch is a more complex sugar made of strands of glucose molecules joined together. These are stored in the leaf and other parts of the plant such as the _____ (e.g. apples), _____ (e.g. rice), _____ (e.g. potatoes) and _____ (e.g. carrots). These starches are food to animals, including humans.



Starch is made of a strand of glucose

Glucose is also converted into cellulose. Every cell in a plant is surrounded by a cell wall made of cellulose. Cellulose is a structural material that provides support for the plants so that they can grow tall without falling over. Cellulose is what provides fibre in our diet. (Fibre cannot be broken down in our bodies, but help in healthy digestion.)



Plant cell wall

Glucose, starch and cellulose are all types of _____. Glucose is a simple carbohydrate, while cellulose is a complex carbohydrate.

Respiration

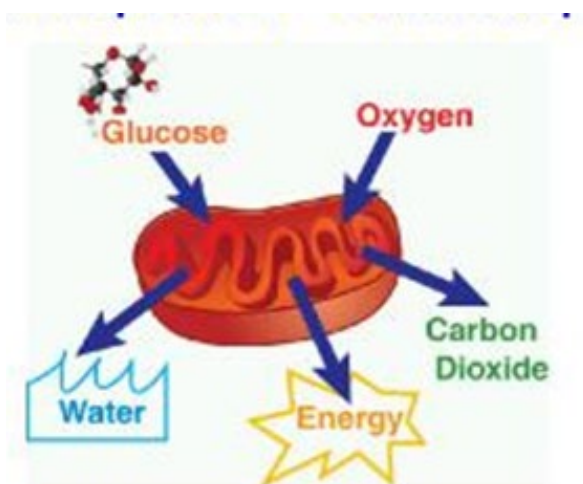
Respiration releases energy from food

Living organisms require _____ to survive. In the process of _____, food is made by the plant. This food contains _____ energy. This energy can then be released from the _____ in a series of _____. In the process of respiration, the glucose is broken down inside cells to release energy.

Note that although it is only plants that photosynthesise, all living organisms respire. Although plants make their own food, they also need to break down glucose to release energy when energy is needed. Respiration takes place in every _____ of every living organism.

Oxygen is required for the process of respiration. Plants absorb oxygen from _____. Animals _____ the oxygen. During the process of respiration, plants and animals require glucose and oxygen to produce energy, carbon dioxide and water. The energy is used by the living organism. The carbon dioxide and water are both _____. Plants release the carbon dioxide and water via their _____. Animals _____t the carbon dioxide and water.

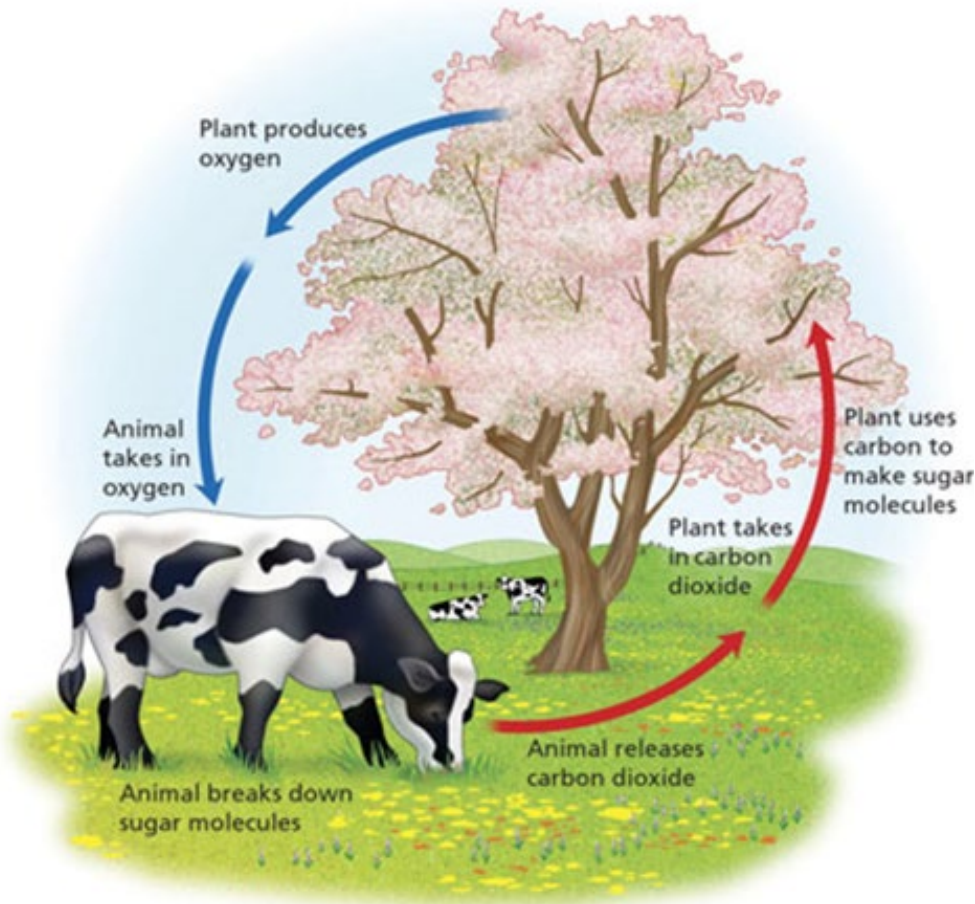
The chemical reaction for respiration is:



The process of respiration in the cell

Respiration and photosynthesis cycle

Look at the diagram below, which shows the cycle of respiration and photosynthesis.

**Respiration and photosynthesis cycle**

The cow eats the grass. Grass contains the _____ stored after photosynthesis. The grass is digested by the cow and broken down into _____ molecules. The glucose molecules are transported by the cow's bloodstream to every _____ in the cow's body, where _____ takes place. The cow uses the _____ that is released by the respiration process for its _____ (e.g. _____).

Respiration and photosynthesis may be thought of as opposite processes.

- During photosynthesis, plants use energy from the Sun to produce food for themselves and for animals. _____ is used and _____ is released.
- During respiration, food is broken down and energy is released. _____ is used and _____ is released.

Conducting investigations

Every investigation is set out as follows:

Observation and research question

We start off by being curious about how nature works. That leads us to ask questions. For example, let's say you have a plant growing in your garden in full sunlight. This plant is large and healthy. You have the same plant species growing under a tree in your garden, and that plant does not look quite as vigorous. You might then start to question why a plant growing in the shade is smaller than a plant growing in the sun. You may then develop a research question:

Introduction

Scientists investigate and report the results of their investigations so that other scientists can share research. We build new research on previous research. Before designing an experiment, a scientist must research previous findings to see whether the work has already been researched. It is important to give credit to previous research by referencing any research. For example:

Plants make their own food during the process of photosynthesis (reference). When food molecules are broken down in the cells of plants during respiration, energy is produced (reference). It would follow that the more photosynthesis occurs, the more food is made, and the more energy the plant has for growth. Plants seem to grow more in full sun than in shade, so it would seem that the level of light intensity could play a role in the amount of food produced by photosynthesis.

Aim

Next, you outline your aim. This is a statement declaring the intentions of the investigation. It is linked to the hypothesis.

Hypothesis

You now need to develop a hypothesis. The hypothesis is the predicted answer to the research question. It is written as a statement, not as a question.

The hypothesis will be written in an if ... then format, for example:

Every experiment has a dependent and an independent variable. Both the dependent and the independent variable appear in the hypothesis.

The independent variable is the **cause** and is not dependent on anything.

The dependent variable is the **effect** and is dependent on the independent variable.

In our experiment, what is the independent variable? Explain.

What is the dependent variable? Explain.

Materials

In the materials section, you list all the materials that you need in order to conduct your experiment.

What materials will you need for this experiment?

Methods

In the methods section, you give step-by-step instructions on how to conduct the experiment. Provide pictures or diagrams where relevant.

The purpose of an investigation that uses the scientific method is to gather EVIDENCE. This evidence is gathered in the form of MEASUREMENTS that are objective (not

opinions that are subjective).

The reliability of the evidence depends on how much evidence the scientist gathers.

How will we make sure that the test is reliable?

The way the evidence is gathered determines the validity of the experiment.

How will we make sure that the test is valid?

For example, in the experiment we use the same-sized pots, the same type of soil, the same plant species, and we provide each plant with the same amount of water.

We also need to set up a control.

In the experiment, all the variables are controlled. All of the variables are kept the same, but only one is changed (i.e. _____). This means that if there is a change in the dependent variable (i.e. _____), then it has to be caused by the independent variable (i.e. _____).

We therefore set up a control. The reason for the control is to make sure that the independent variable (i.e. _____) is causing a change in the dependent variable (i.e. _____). For the control, _____.

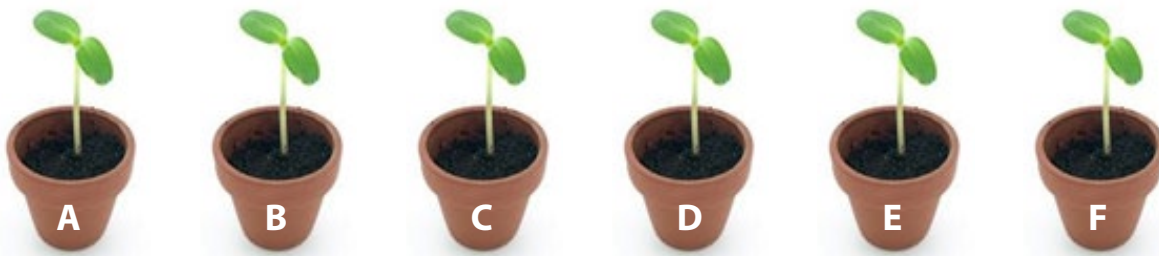
The control is set up with a known outcome. We expect the plant will not to grow in the dark. If the results of the experiment and control are the same (i.e. the plant grows in the light and in the dark), then the independent variable we have chosen is NOT causing the dependent variable to change, and so we must redesign the experiment.

This is an example of how to present the methods:

Place an equal amount of soil into each container. Plant a seedling of the same size and species into the soil in each container.

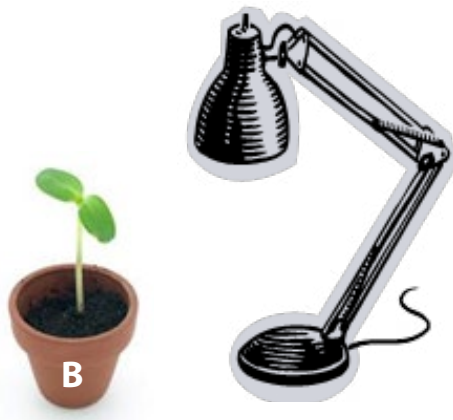


Label the containers A to F.



Place container A in a dark cupboard. This will be the control.

Place container B in a dark cupboard with a lamp directly above the plant.



Place container C in a dark cupboard with a lamp 1 m away from the plant.

Place container D in a dark cupboard with a lamp 2 m away from the plant.

Place container E in a dark cupboard with a lamp 3 m away from the plant.

Place container F in a dark cupboard with a lamp 4 m away from the plant.

Water all the seedlings daily with three tablespoons of water.

Measure the height of the seedlings using a ruler every day for 14 days.

Results

In the results section, we record the observations. We usually tabulate the data, and draw a graph if possible. This makes the results easier to analyse.

Days	Average height of the seedlings in mm					
	Container A	Container B	Container C	Container D	Container E	Container F
1						
2						
3						
...						
...						
13						
14						

Discussion

In the discussion section, you explain your observations in words and explain what your observations tell you. You have to be able to discuss possible errors by discussing reliability and validity.

Can you see any reliability issues with this experiment? Explain.

Can you see any validity issues with this experiment? Explain.

Conclusion

In the conclusion section, you say whether you reject or accept your hypothesis based on your results. You simply say

"I accept the hypothesis that if light intensity increases, then the rate of photosynthesis will increase" OR "I reject the hypothesis that if light intensity increases, then the rate of photosynthesis will increase".

References

You need to reference the sources that you used to get all the background information. There are many ways to reference, but you will be using the Harvard method:

In-text referencing

Use the author's surname and year of publication, and include a page number if you are quoting an author. For example, in your introduction you say:

Plants make their own food during the process of photosynthesis (Smith, 1992:250). When food molecules are broken down in the cells of plants during respiration, energy is produced (Johnson, 2001:12).

Where there are two authors, reference as follows: (Smith & Johnson, 1983:145).

Where there are more than two authors, write out all author surnames in the first reference to the source. In subsequent references, write as follows: (Smith et al., 2009:356).

Remember that you have to give a reference for every statement that you make that you sourced from somewhere. Also remember that plagiarism is not allowed. This means that you cannot copy word for word the information that you find in a source. You have to write the information in your own words.

The references list

Every reference that you refer to in the text needs to be shown in full in the reference list.

You would reference a text book like this:

Author, A.A. 1994. Title of work. Location: Publisher.

You would reference an article like this:

Author, A.A., Author, B.B. & Author, C.C. 1994. Title of article. Title of Periodical, volume(issue):first–last page.

You would reference an internet article like this:

Author, A.A., Author, B.B. & Author, C.C. 1994. Title of article. Title of Periodical, volume(issue):first–last page. Retrieved from web address [Accessed day month year].

You would reference an internet website like this:

Author, A.A. 2000. Title of work. Retrieved from web address [Accessed day month year].

Assignment 2: Conducting an investigation

We will be looking at the research question: **Does the amount of water that a plant gets affect the growth of the plant?**

You will work in pairs, but you will present your own investigation report, which must include the following:

- Introduction
- Aim
- Hypothesis
- Materials
- Methods (make sure that your experiment is valid and reliable. Include pictures and/or photos)
- Results (to include a table and a graph)
- Discussion
- Conclusion
- References

The report may be typed or handwritten.

Your investigation will be assessed as follows:

Criteria	Achieved	Partially achieved	Not achieved
The investigation was set out correctly.	All the headings present and in the correct order(2)	Not all the headings present / not in the correct order (1)	Inadequate setting out of the report (0)
Introduction	The introduction was relevant and well written (6)	The introduction was not entirely relevant (3)	The introduction was not outlined correctly (0)
Aim	The aim was outlined correctly (2)	The aim was outlined in its entirety (1)	The aim was not outlined correctly (0)
Hypothesis	The hypothesis was outlined correctly (2)	The hypothesis was outlined incorrectly (1)	The hypothesis was not outlined (0)
Materials	All the materials were listed (2)	Not all the materials listed (1)	The materials were not listed correctly (0)

Methods	The methodology was correctly outlined with all the relevant detail and pictures. The methodology was valid and reliable. (6)	The methodology was not outlined with all the relevant detail and pictures. The methodology was not entirely valid and reliable. (3)	The methodology was not adequately outlined. The methodology was not valid and reliable. (0)
Results	The results were collected, tabulated and graphed correctly (10)	The results were not presented in their entirety (5)	The results were not presented correctly (0)
Discussion	The discussion was relevant and well written (6)	The discussion was not entirely relevant (3)	The discussion was not outlined correctly (0)
Conclusion	The conclusion was outlined correctly (2)	The conclusion was outlined incorrectly (1)	The conclusion was not outlined (0)
References	Done correctly in the text and in the reference list (6)	Completed but not entirely correctly (3)	Not done in text/reference list (0)
Neatness of presentation	The presentation was neat (2)	The presentation was adequate (1)	The presentation was inadequate (0)
Peer review	Highly involved (4)	Partly involved (2)	Not involved (0)
TOTAL 50			

INTERACTION AND INTERDEPENDENCE WITHIN THE ENVIRONMENT

Introduction

Plants and animals live together and interact in ecosystems. They interact with one another and with non-living components in the ecosystem.

Organisms rely on their ecosystem to satisfy their need for shelter, food and space for breeding and for hiding from predators.

There are different types of relationships between living organisms in an ecosystem. What is the most common relationship between living organisms?

What are some other types of relationships?

Provide three examples of interactions between living organisms

Provide three examples of interactions between living organisms and their non-living environment.

Ecology

What is ecology?

What are living organisms?

Living organisms have been divided into five kingdoms: bacteria, protists, fungi, plants and animals. We will learn more about bacteria, protists and fungi later.



The five kingdoms

What are some examples of the non-living components?

Interactions between living things

When scientists study living things in an ecosystem, they classify the interactions at different levels:

- Populations
- Communities
- Ecosystems
- Biosphere

What is a population?

Consider the diagram below. There are two prides of lion in the south of the Kruger National Park and one pride of lion in the north of the Kruger National Park. Is one pride a population? Do all three prides make a population? Do two prides make a population? Explain.



In the north of the Kruger National Park



In the south of the Kruger National Park



To be a population, the individuals of the same species need to interact with one another.

What is a community?

For example, in a wetland community, there may be a population of fish, a population of frogs, a population of algae and a population of reeds.

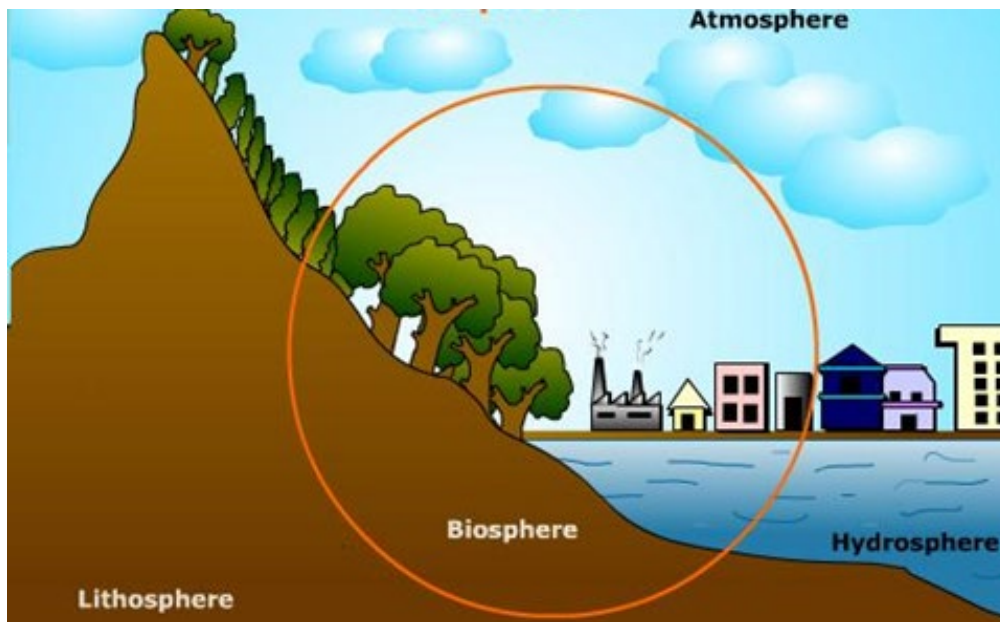


Wetland

What is an ecosystem?

In an ecosystem there are populations and communities. There are also _____ where organisms live.

What is the biosphere?



The biosphere consists of the lithosphere, hydrosphere and atmosphere

Ecosystems

There are many different types of ecosystems found on land and in water. Examples of ecosystems are: _____.

All of the Earth's ecosystems make up the _____.

The non-living parts of the ecosystem influence which living organisms can live there. For example, in a specific temperature and soil type, certain types of plants can thrive. Based on the plants that grow in an area, certain animals will be able to thrive.

The living and non-living parts of the ecosystem interact with one another.

Biotic and abiotic factors

An ecosystem consists of two parts:

- The _____, known as the biotic factors
- The _____, known as the abiotic factors.

Biotic factors include all the living organisms that interact with one another. They feed on one another, compete with one another and benefit from one another. For example, zebras and impalas compete for food; plants compete with one another for light; male giraffes will fight for a mate; barnacles on a rock compete for space.



Barnacles on a rock compete for space



Giraffes fighting for a mate

Abiotic factors are the non-living component of the ecosystem. Abiotic factors include, among other things:

The table below shows how each one of these abiotic factors affects the biotic environment:

Temperature	Most organisms on Earth live in temperatures of between 0 °C and 40 °C. Each organism grows most optimally at a certain temperature range. For example, a yak is adapted to living in very cold temperatures and a camel is adapted to living in very hot temperatures. Temperature varies with season as well as the time of day.
Wind	Wind can stunt the growth of plants and affect the migration patterns of birds. Wind is also important in pollination and seed dispersal.
Water	Water is essential for the survival of living organisms. Some plants and animals are aquatic or semi-aquatic, meaning that they live in water. In ecosystems where water is scarce, plants and animals need to be able to store water and reduce their water loss. Water cycles through the biosphere in the water cycle.
Light intensity	Plants need light in order to photosynthesise. Some plants grow better in shade and other plants grow better in full light. The amount of sunlight received during the day will vary with the time of day and with seasons.

Soil	There are different soil types, namely sand, loam and clay. Each type of soil has different properties, such as the amount of nutrients that it can hold. These properties influence which plants can grow in them.
Slope	Slope describes the steepness of the land. When a slope is steep, it is less likely to support plant life because water runs off it quickly and there is more soil erosion.

The size of an ecosystem

The size of an ecosystem varies. It needs to be defined by the scientist that is studying the particular ecosystem. An ecosystem may be large, such as the entire savanna biome in South Africa, or the Knysna forest. An ecosystem may be small, such as a rotting log, a puddle of water, or your back garden.

Survival in an ecosystem

For plants and animals to survive in an ecosystem, they need to be able to cope with changes to their habitat. Such changes may be gradual or sudden.

What are examples of gradual changes?

What are examples of sudden changes?

Some organisms are able to cope by adapting quickly to these changes. Why are some organisms able to adapt quickly to changes?

Rats, starlings and weeds are examples of species that are able to survive in many ecosystems and survive well when the ecosystem changes.

**Rat****Starling****Weed**

Feeding relationships

We may classify living organisms into groups according to what they eat. This helps us to determine feeding relationships within an ecosystem.

Producers

What are producers?

Plants trap sunlight and use it to make food during the process of photosynthesis. Some protists, such as algae and seaweed, are also able to photosynthesise. Some of this food is used by the plant/protist and some of it is stored as _____.

Plants are essential to animals as a food source. Plants are also important to people for other reasons. For example, the marula tree is indigenous to South Africa. The marula tree is very important to local communities. These trees produce a yellow fruit the size of a plum in large numbers. The fruit and seeds are eaten; oil can be derived from the seeds; the bark is used to treat malaria, stings and bites; the leaves are used to treat heartburn; and ink is made from the gum.

**Marula tree****Marula fruit**

Baobab trees occur in South Africa and are also important to local communities. The fruit is eaten and is highly rehydrating, has medicinal value and is used to alleviate stomach aches and fevers.



Baobab tree



Baobab fruit

Consumers

What are consumers?

Animals, fungi, bacteria and some protists cannot make their own food.

Some animals get food by eating plants. They are known as _____. Examples include cows, elephants, giraffes, sparrows, locusts and aphids.

Some animals get food by eating other animals. They are known as _____. There are three types of carnivores: predators, scavengers and insectivores.

Predators are carnivores that _____. The animals being hunted are known as _____. Examples of predators include lions (with zebras and wildebeest as examples of prey) and eagles (with meerkats and lizards as examples of prey).

Scavengers are carnivores that _____. Examples of scavengers include vultures, hyenas, wild dogs and jackals.

Some predators, such as lions, will scavenge if the opportunity arises. Some scavengers, such as hyenas and wild dogs, also hunt and kill their own prey.

Insectivores are carnivores that _____. For example, ladybirds feed on aphids and small insects, the anteater feeds on ants, and the aardwolf eats termites.



Anteater



Aardwolf

Some animals get food by eating both plants and animals. They are known as _____. Examples of omnivores include humans, warthogs and baboons. Many bird species eat insects and plant materials and are considered omnivores.

Decomposers

What are decomposers?

Examples of decomposers include bacteria, fungi, beetles and earthworms. Decomposers vary from being microscopic to macroscopic.

Decomposers digest and in turn break down dead matter and waste into simpler substances, which are then released into the soil. These simpler substances are _____ that can then be taken up by _____. In this way, nutrients are recycled in the ecosystem by decomposers.

For example, dung beetles are decomposers that feed on the dung of large herbivores. They break down the large parts of dung as they collect and feed on the dung. This broken-down dung mixes with the soil and enriches it with nutrients (like fertiliser). This improves plant growth.

**Dung beetle**

Fungi are decomposers that digest food outside of their body. They release digestive enzymes onto the dead matter. The digestive enzymes break down the dead matter into simpler nutrients. The fungi take up some of these nutrients, but some remain in the soil. Examples of fungi are mushrooms and mould.

**Mushrooms****Mould**

Decomposers are essential in an ecosystem. Why?

Without decomposers, soil nutrients would eventually run out and plants would not be able to grow. Decomposers are also important to people, as they are used to treat human sewage. Sewage contains human waste, bits of food and chemicals. Bacteria are used to break down the sewage into simpler substances. This purifies our water.

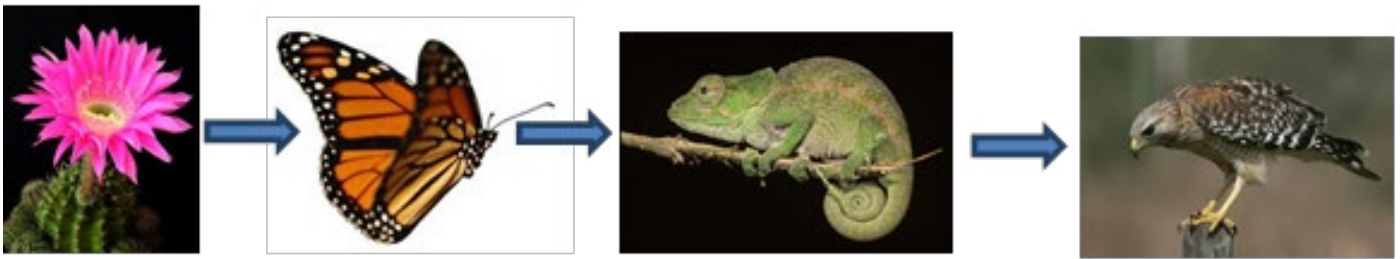
Food chains, food webs and energy pyramids

Plants and photosynthetic protists such as algae play a crucial role in the ecosystem. They are _____. They capture the light energy from the Sun and use this energy to produce food in the process of photosynthesis. This energy is passed along a food chain in an ecosystem.

Food chains

A food chain represents a feeding relationship between living organisms. A food chain will always start with a producer, because they can make their own food. A food

chain may end with decomposers that return nutrients to the soil. Decomposers are not always shown in simple food chains such as the one below:



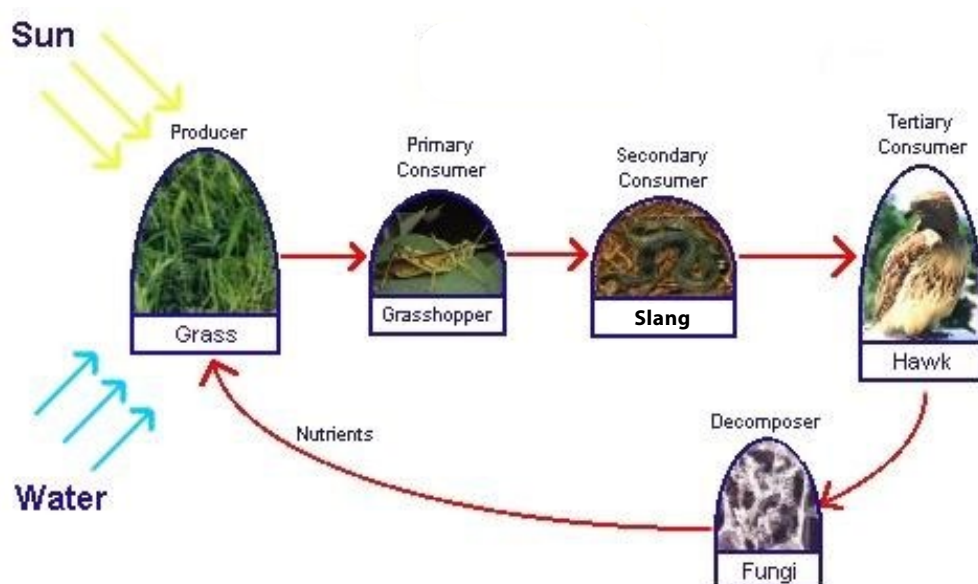
What does this food chain show?

The _____ eats the _____. The _____ eats the _____. The _____ eats the _____.

The flower is a _____. The butterfly, chameleon and hawk are _____. Consumers are named based on their position in the food chain:

- Primary consumer: This is the first consumer in the food chain that feeds on a producer. Example: _____.
- Secondary consumer: This is the second consumer in the food chain and feeds on the primary consumer. Example: _____.
- Tertiary consumer: This is the third consumer in the food chain and feeds on the secondary consumer. Example: _____.
- A food chain may include a quaternary (fourth level) consumer, and may even include higher levels. We will only go up to quaternary consumers.

Consider the food chain below:



Producers get energy from the _____ to _____. The _____ eats the _____. The _____ eats the _____. The _____ eats the _____. When the hawk dies, it gets _____ by _____. The nutrients from the hawk are then returned to the _____ to be taken up by the _____.

What is the difference between the first and the second food chains above?

A food chain is the feeding of organisms on one another in a sequence. What does it show?

What do the arrows represent?

Energy is passed through an ecosystem along a food chain from the producers to the consumers. When an impala (_____) eats grass (_____), energy is transferred from the _____. When a lion (_____) eats an impala, energy is transferred from _____. Decomposers are the last link in the transfer of energy when they break down dead plants and animals in each link in the food chain.

A food chain in an ecosystem ends with a top predator – an animal with no natural enemies, such as an alligator, polar bear or elephant.

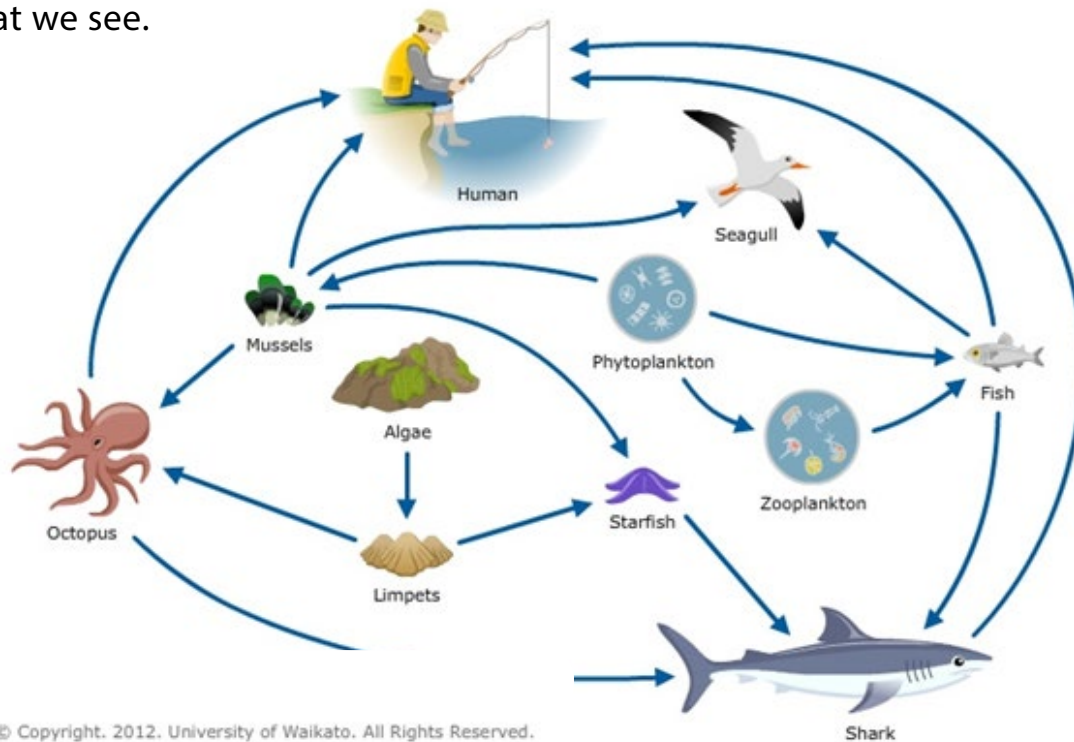
Each link in the food chain relies on the one before. The links are all dependent on one another. What will happen if one link is removed?

Remember that food chains are simple feeding relationships. You will see now how this is different from a more complex food web.

Food webs

In any habitat there are many different food chains that are cross-linked. This is because most animals do not eat only one kind of food. This cross-linking of food chains is known as a food web. This is a more complex feeding relationship than the food chain, because it shows many relationships that go on at the same time.

See the diagram below on the food web in a marine environment. Let's explain some of what we see.



In the case of the producers, there will not be an arrow coming towards it, only going away from it. Why?

Which are the two producers in this food web?

Give three examples of primary consumers in this food web.

What does the fish eat?

What eats the fish?

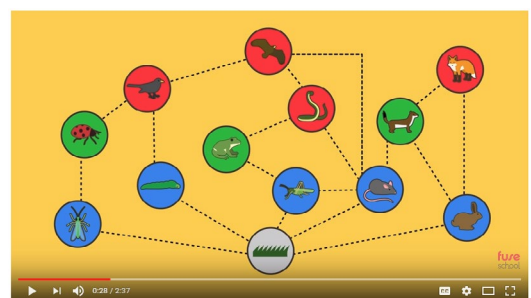
Is the octopus a primary, secondary, tertiary or quaternary consumer? Provide two examples.

When is the shark a secondary consumer? Explain.

When is the shark a tertiary consumer? Explain.

The human is a quaternary consumer when it eats the shark. Explain.

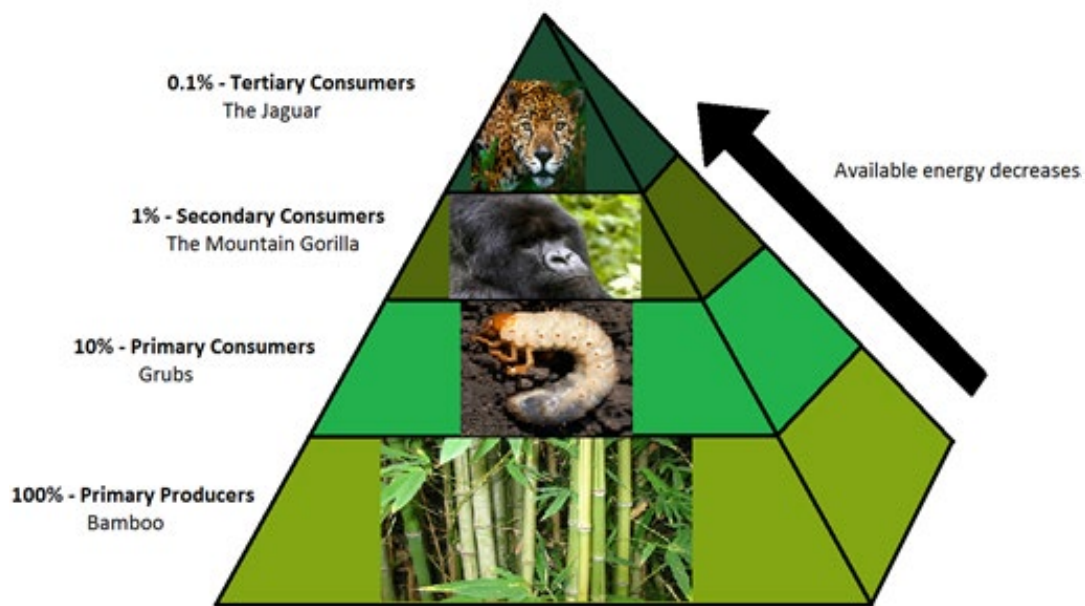
<https://www.youtube.com>



Energy pyramids

Each step in the food chain or food web is called a trophic level. The first trophic level is made of _____. The second trophic level is made of _____. The third trophic level is made of _____ and so on. Each consumer depends on the trophic level below it for energy.

See the energy pyramid below that shows the different trophic levels in a typical ecosystem. Explain what you see. Note that in an energy pyramid we do not include decomposers in a trophic level, but they feed at each level.



The _____ eats the _____. The _____ eats the _____.
The _____ eats the _____.

This is very similar to a food chain. However, there is a lot of significance in the pyramid shape. At which level is there the most energy and mass?

When the grub eats the bamboo, only a fraction of the energy that the grub gets from the bamboo becomes new body mass in the grub (_____). The rest of the energy is used by the grub in _____.

When the gorilla eats the grub, the grub passes only a small amount of total energy that it contains in its body for the gorilla to grow (1%). The rest of the energy will be used up by the gorilla in its life processes.

Energy gets lost at each link, and therefore we cannot have the same mass of plants as lions. The total mass of producers will always be _____ than the total mass of consumers. The total mass of primary consumers will always be _____ than the total mass of secondary consumers. It is not about the size or the number of the individual organisms, but about the total mass of the population in the ecosystem.

Balance in an ecosystem

In an ecosystem, the living organisms and the non-living environment that supports them are in fine balance. The number of organisms that an ecosystem can support depends on the resources available.

What are resources?

If there are lots of resources in an ecosystem, it can support _____ different types of individuals and _____ populations. For example, in a _____ there are many resources.

If there are few resources in an ecosystem, it can support only a _____ types of individuals and _____ populations. For example, in a _____ there are few resources.

If food is in short supply, animals will compete with one another for the food. The _____ animals will survive. If soil nutrients are in short supply, plants will compete with one another for the nutrients. The _____ plants will survive. When there is a shortage of food, e.g. if there is a drought, the number of organisms will _____. When there is a lot of rain and plants are plentiful, the number of organisms will _____.

All living organisms in an ecosystem interact with one another. What happens to one population affects the whole community. For example, if there is a lot of rain, lots of berry bushes can grow.

So, the population of mice (that eat the berries) will _____. In turn, the population of owls (that eat the mice) will _____.

If rain is scarce, few berry bushes can grow. So, the population of mice (that eat the berries) will _____. In turn, the population of owls (that eat the mice) will _____.

There is a balance between predators and prey. The size of the populations changes so that they are in balance. If an ecosystem does not remain in balance, it will fail.

The balance in an ecosystem can be disrupted by:

- Natural factors
- Human factors.

What does it mean when an ecosystem is disrupted?

Natural factors that disrupt ecosystem balance

Sometimes there are large disruptions to the ecosystem so that it is unable to cope. A natural disruption would be a natural cause that results in an interference with the normal process of an ecosystem.

Ecosystems may recover from major disruptions, but if the disruption is severe, the damage may be so bad that the ecosystem cannot recover and living organisms may be permanently lost from the ecosystem.

Natural disruptions include things such as _____.

Human factors that disrupt ecosystem balance

Pollution

When humans create pollution in the environment, they release toxic substances into the environment.

What are some common pollutants?



Pollution from factories



Pollution from cars.



Fertiliser running off into a water body

Farmers add large quantities of fertiliser to soil. Fertiliser contains nitrogen, sulphate and phosphate, which are plant nutrients and help crops to grow well. However, when it rains, a lot of these nutrients get washed into rivers. This pollutes the water and causes harm to aquatic organisms.

Above , it was already mentioned that harmful gases are released by power stations when _____ are burnt to make electricity.

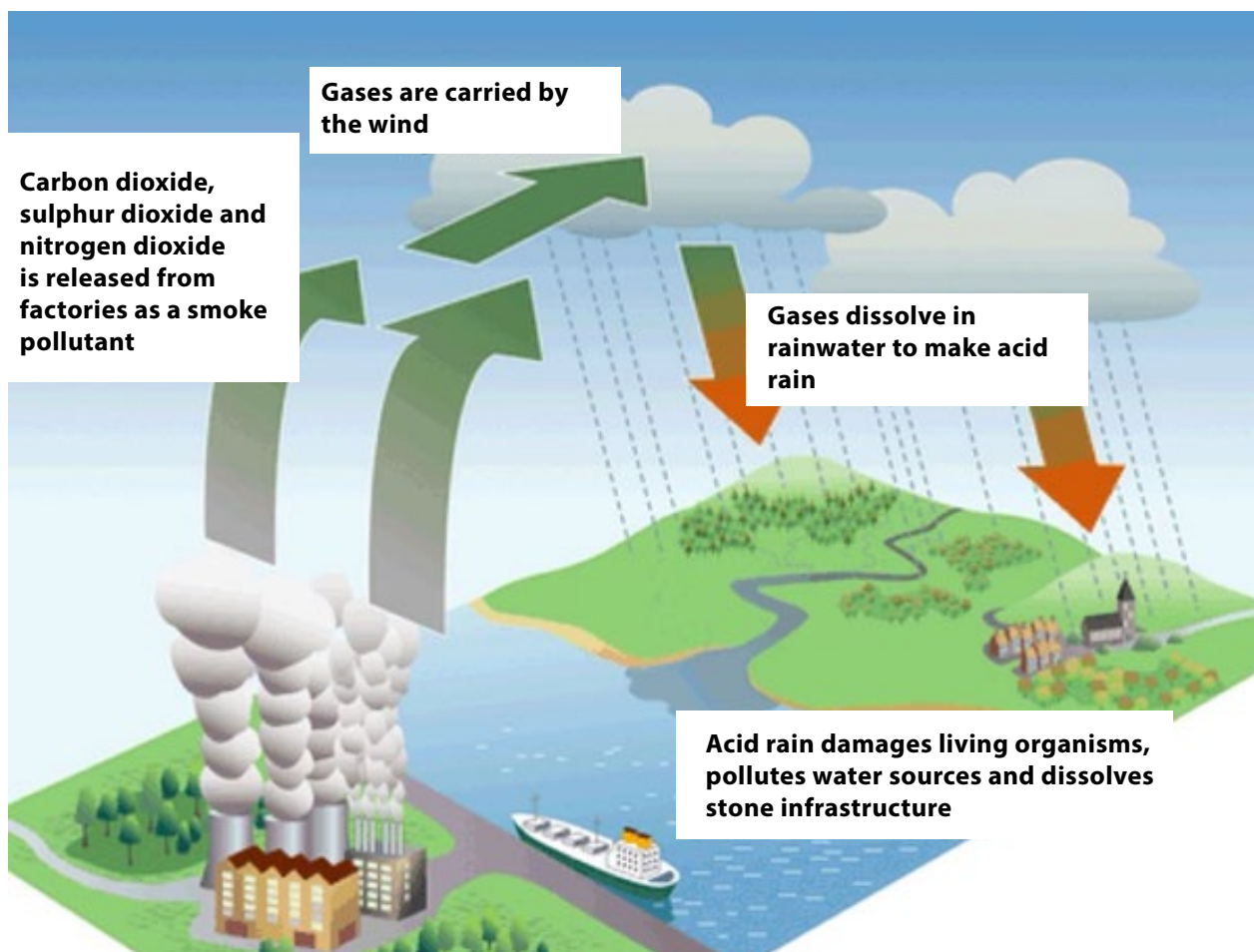
Three of these harmful gases are carbon dioxide, sulphur dioxide and nitrogen dioxide. These react with water when it rains to produce carbonic acid, sulphuric acid and nitric acid. This results in acid rain. Acid rain can damage crops, it can change the pH of rivers and lakes and in turn damage aquatic life and the animals that drink the water. Some aquatic species cannot survive the acidic conditions. Many aquatic organisms die and the biodiversity of the water decreases. Acid rain also damages infrastructure, especially when made of stone.



Leaf affected by acid rain.



Damage to a statue due to acid rain.



Process of acid rain.

Factories often release outflows of warm water into rivers. The increase in water temperatures is known as thermal pollution. It results in less oxygen content in the water, which many aquatic organisms cannot tolerate. Many aquatic organisms die and the biodiversity of the water decreases.



Sugar factory releasing warm water into a river.

Poaching

Poaching is the illegal removal of animals and plants from their habitat. Many of South Africa's plants and animals are under threat from poaching. Species of orchid, cycad, tortoises, lizards and abalone (a mollusc considered a delicacy in certain countries, especially Chile) are taken or caught and illegally exported to other countries where they can be sold for a lot of money. Rhinos and elephants are killed for their horns or tusks and exported, mainly to Asia.



Orchid



Cycad



Abalone

If poaching continues at a high rate and for a long time, eventually the population reaches a point where it cannot recover, and populations die out. In severe cases, the entire species may even become extinct.

Think of this as a full biscuit jar in a room full of hungry people. The more people that take biscuits, the fewer are left in the jar. It takes time to bake more biscuits and if the biscuits in the jar are eaten faster than they can be made, soon you run out of biscuits. This is an analogy to understand why eventually a population may get too small to recover.

This can affect the ecosystem balance. For example, if a carnivore population is removed from the ecosystem, there will be an _____ in the herbivore population, as it lost a predator. This will result in a higher demand for the plants that the herbivores eat and the plant population is likely to _____.

Habitat destruction

One of the biggest man-made threats to ecosystem balance is the destruction of habitat. For example, large portions of the Amazon have been cut down to make way for crops and cattle farming. Firstly, the size of the ecosystem decreases, so competition for resources _____.

. Some living organisms cannot adjust to the loss of habitat. Why?

Secondly, habitat destruction changes water patterns and availability, which affects all living organisms. This changes the entire ecosystem balance in very complex ways. A portion of the Amazon jungle in Brazil that has been cut down. There is a single tree growing in the field, because ironically it is illegal to cut down the Brazil nut tree.



Invasion of alien species

Plants, animals and other living things may intentionally or accidentally be introduced by humans to

a place where they have not lived before. They become invasive species when they out-compete the native (naturally occurring) species. They may be bigger, stronger, faster-growing or more aggressive than the native species.

If they are, then they spread quickly and take over an ecosystem. The native species are often driven out or killed. Invasive species upset the balance of an ecosystem.

For example, jacaranda trees are native to north-eastern Argentina. The tree was brought to South Africa in 1888, where the tree was planted in Pretoria due to its beauty. People liked the look of it so much that they planted it all over Pretoria, to the extent that it became known as the Jacaranda City.

This plant has adapted well to warmer parts of the country. It is very resilient. It can survive drought and heat. It is believed that jacaranda trees are water-hungry and out-compete native species in the fight for water, eliminating native species from areas where they previously grew naturally. The papery, winged seeds also disperse easily and over large distances.



Jacaranda trees lining the streets of Pretoria

One example is the overharvesting of fish. Humans have caused the population decline of hundreds of fish species by overfishing or overharvesting them.

The bluefin tuna is a predator in the ocean. It eats small fish and invertebrates such as sardines, herrings, squids and crustaceans. Thereby, these populations are kept under control. Remember

that ecosystems work in balance; one small change can have drastic effects for the ecosystem as a whole. For example, we said that the bluefin tuna eats small fish. These small fish eat algae and other small plants.



Overharvesting of fish in a village

Overharvesting

Humans often take too many of one species from their natural habitat. Typically, this involves a species used as a food source. When a species is harvested in large numbers and at a fast rate, the population gets too small and cannot recover. It is then considered overharvested. Remember our biscuit jar analogy.



Leaves, flowers and seeds of the jacaranda



Bluefin tuna

GRADE 8	TERM 1	NATURAL SCIENCES MODULE 1 LIFE AND LIVING	
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If the bluefin tuna were extinct, then the number of small fish would drastically _____. This means these fish would need to eat _____ algae, which is a producer and important to maintain the ecosystem. This means that hundreds of species of aquatic plants and animals would die due to the lack of producers in the ecosystem.

ADAPTATIONS

Defining adaptation

What is an adaptation?

The polar bear lives in the coldest place on earth, the Arctic. The polar bear has very thick fur. The hairs are hollow and they trap and warm the air. The skin of the polar bear is black to absorb more light and warm up the skin. It also has many layers of fat under its skin. This keeps the polar bear warm. Is this a behavioural or physical adaptation? _____

While the polar bear is swimming, it actually stays dry with the help of guard hairs. Guard hairs maintain a layer of dry air next to the skin and repel liquid water. This keeps the polar bear dry and warm while it is swimming. Is this a behavioural or physical adaptation? _____

Some animals are nocturnal, which means that they hunt for food at night. They have adapted as such because there is less competition at night. Is this a behavioural or physical adaptation? _____

Some frogs hibernate in winter by burying themselves in the mud. This means that they 'sleep' through cold weather. They do this because in cold weather their body temperature would drop too low for them to survive. When the weather warms up again, their body returns to a temperature at which they can function. Is this a behavioural or physical adaptation? _____

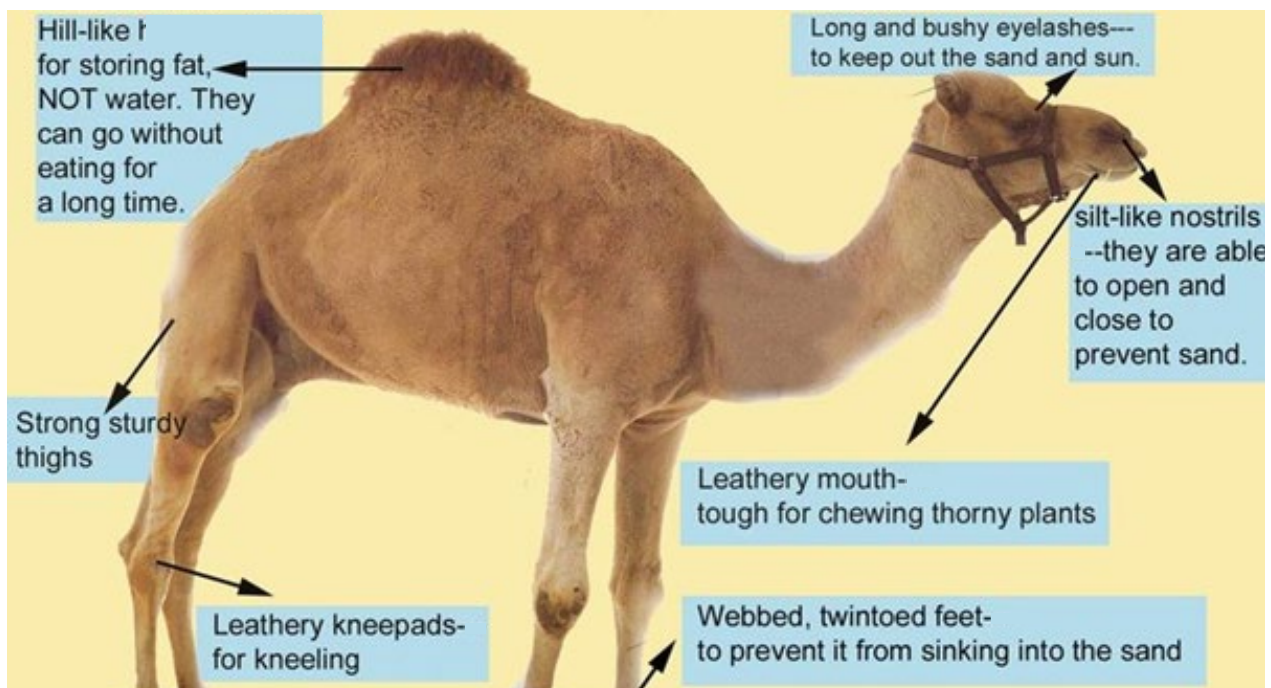
Cheetahs are predators. They have good eyesight and can run very fast. They have sharp teeth and claws to hold onto the prey and to tear into the flesh of the prey. Is this a behavioural or physical adaptation? _____

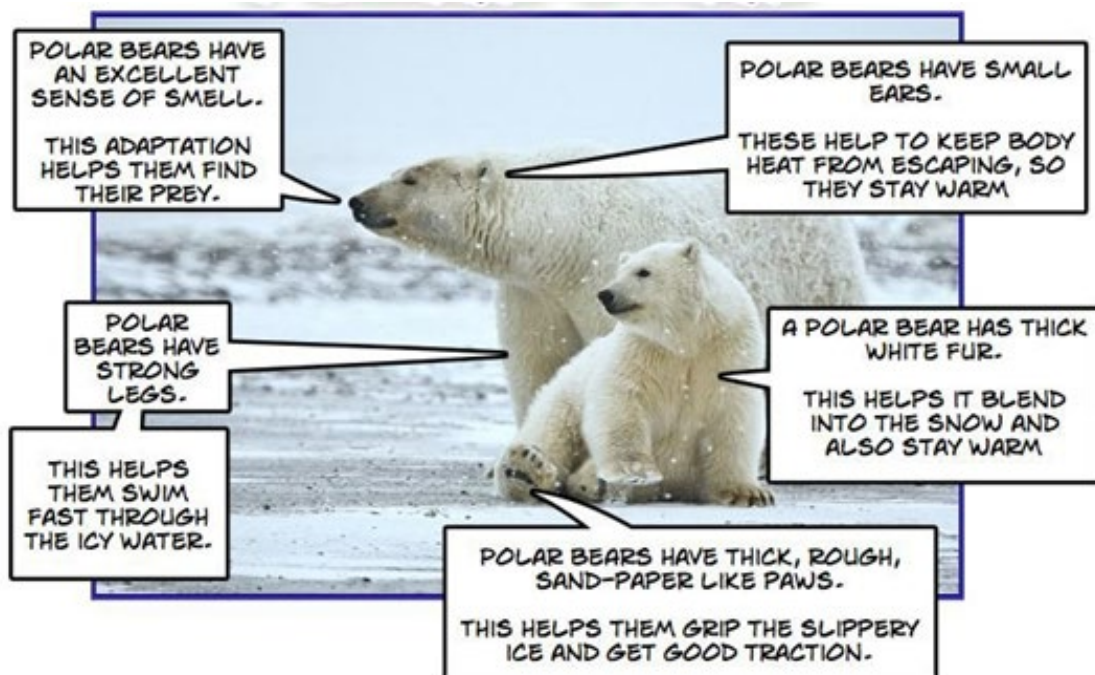
Cheetahs are well camouflaged, meaning that they blend in well with their environment. Is this a behavioural or physical adaptation? _____

Sharks are also predators. They have a good sense of smell and are fast swimmers to catch their prey. They have a muscular tail to propel them through the water. Is this a behavioural or physical adaptation? _____

Sharks hold their prey with sharp teeth and shake their head from side to side to tear it into pieces. Is this a behavioural or physical adaptation?

In extreme environments, living organisms display more characteristic adaptations. Here are the camel and polar bear as examples:





Assignment 3: Adaptation

Research a living organism and provide at least five adaptations of this organism that help it to survive in its environment. For each adaptation, say whether it is a behavioural or a physical adaptation.

You cannot choose a tree or a dolphin. This is too broad. You need to choose an actual species. For example, for a tree you need to choose a white pine tree or a scarlet oak tree. For a dolphin you may choose the freshwater dolphin or the bottlenose dolphin.

You will present this to the class and then hand it in. You will also include references.

Changing environmental conditions and adaptations

The conditions in the environment of living organisms may change. For example, the environment may become hotter and drier, or there may be a shortage of resources. Living things that are able to adapt well are _____. The living organisms that are best adapted will have a better chance of getting the _____ such as food, water and shelter. This will allow them to _____.

Living things adapt by chance. Adaptations are passed on from the parents to the offspring. These differences are known as _____.

Humans have variations. Some of us are shorter and some are taller. Some have brown eyes and some have blue eyes. Often, if we have two tall parents we will also be tall. If both our parents have brown eyes, we are more likely to have brown eyes. In the wild, there are also variations in living organisms that are passed on from parents to offspring.

Look at the picture below. This young zebra was born darker and with very few stripes.



A zebra darker than the rest of the population

The question is: Will this variation help the zebra to survive in the wild? If this variation makes it easier for predators to spot the zebra and eat it, then this zebra will die before it has a chance to pass on its variation to its own offspring.

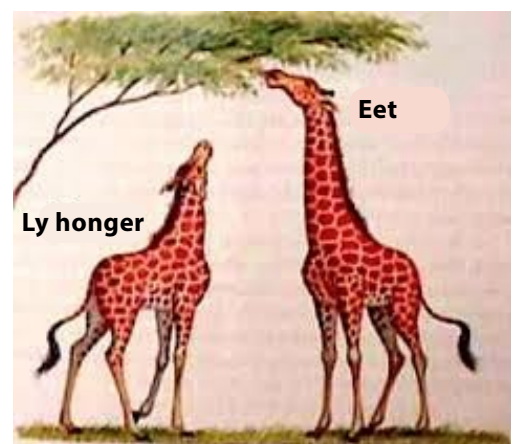
If this variation makes it harder for predators to spot the zebra and eat it, then this zebra is more likely to survive to adulthood, have babies and pass on

its variation to its own offspring. If this variation is really a good one, then over generations, more and more zebras will be born darker and with fewer stripes.

Over many, many years, eventually all the zebras will be darker and with fewer stripes.

(It so happens that this zebra was eaten by hyenas at six months of age. Zebras are sometimes born with this variation, and none of them survive past six months because they are more visible to predators.)

Say a giraffe is born with an especially long neck and another one with an exceptionally short neck. This is not an issue when there is a lot of rain, trees are growing and food is plentiful. But now there is a drought. Food is limited. All the lower leaves have been consumed because they are easier to reach. The short-necked giraffe will starve and die. The long-necked giraffe will survive the drought, and have babies that are more likely to have long necks too.



Lack of adaptation causes extinction

Some organisms are unable to adapt where there is a change in their environment, especially when it is a sudden change. Such organisms may be eliminated from their ecosystem. If there are no more species of that organism in another ecosystem on Earth, then that species will be considered _____.

The dodo was a flightless bird found in Mauritius only. When sailors came to the island, they hunted the dodo for food. The sailors introduced rats and pigs to the island and these animals ate the dodo's eggs and young. The dodos had no way of escaping the predators and there were no dodo populations anywhere else in the world, so the dodo became extinct.



<https://www.youtube.com/watch?v=0hanQEXPjFo>

Adaptations of plants

Some plants are adapted to hot and dry environments. Such plants are called _____.

**Cactus plant****Agave species**

Xerophytes therefore have adaptations that _____

Xerophyte roots are short and widely branched out. They spread out just below the soil. Why?

The roots are thick and fleshy. Why?

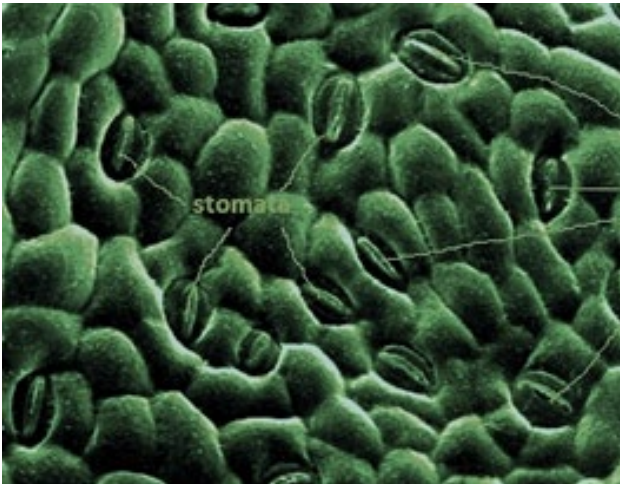
Their stems and leaves are thick and fleshy. Why?

Some plants are adapted to living in water. Such plants are known as _____.

**Water lily**

Their roots tend to be short and poorly developed. Why?

The leaves are large and flat and float on the water. Stomata are tiny holes in the leaves where gasses enter and leave.



Stomata on leaf surface

adaptations that we can learn about.

Hydrophytes have many stomata on the upper surface of the leaves. Why?

Mesophytes are plants that live in moderate conditions. In other words, they cannot grow in areas of extreme drought and excessive water. They get an average and regular supply of water. Like every other living organism, each plant will have its own set of adaptations. However, there are no general

Camouflage and mimicry

In order to survive, predators need to find prey and prey needs to hide from predators. Camouflage and mimicry help animals to hide and stay hidden.

What is camouflage?



For example, this toad has patterns, shapes and colours like the rocks surrounding it.



This stick insect is shaped, patterned and coloured like the leaf that surrounds it.



A chameleon has skin pigmentation that allows it to change colour with its surrounding environment.

In winter, the arctic hare has white fur to match its snowy surroundings, but in summer, when plants are flourishing, the fur turns brown.



Arctic hare in winter.



Arctic hare in summer.



<https://www.youtube.com/watch?v=RBdbGPK1ZIQ&t=2s>

What is mimicry?

The mimic is usually _____.

The animal that is being mimicked (known as the model) is usually _____.

The mimic relies on the fact that _____.

For example, the monarch butterfly has black and orange wings. It tastes bad and birds avoid eating it. The viceroy butterfly tastes good to the birds. However, the birds have learnt that butterflies with black and orange wings should be avoided. The viceroy butterfly is the _____ and the monarch butterfly is the _____. The owl butterfly has huge 'eye' spots on its wings, which resemble owls' eyes. Lizards



Koningvlinder

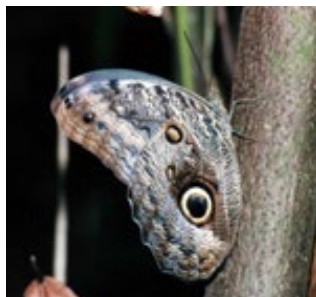


Onderkoningvlinder

and birds that would want to eat the butterfly are mistaken into believing that it is an owl and leave it alone. The butterfly is the _____, the owl is the _____.



Owl butterfly



Owl

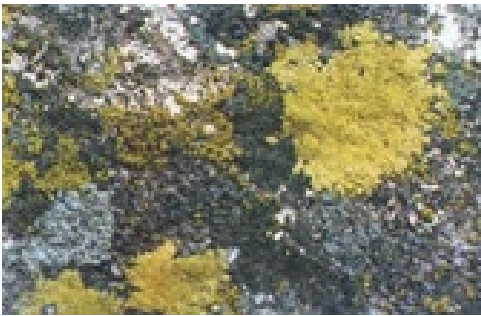
Symbiosis

Living organisms interact with one another all the time. Symbiosis is a type of interaction between living organisms of different species that helps one or both organisms to survive.

There are three symbiotic relationships: mutualism, commensalism and parasitism.

Mutualism

What is mutualism?



Lichen – fungus and algae

Lichen is made up of fungus and algae that grow together.

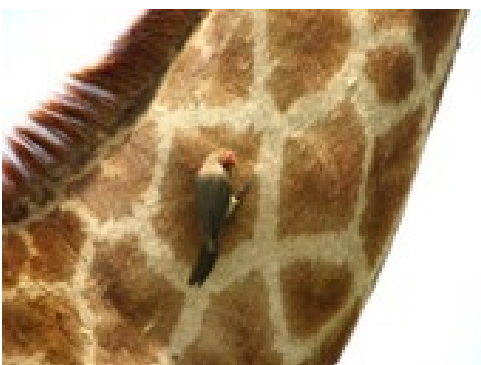
The algae photosynthesise and make food for themselves and for the fungus.

The fungus protects the algae from drying out and shades them from strong sunlight.



Butterflies and flowers

Butterflies get food from pollen in flowers. The flowers get pollinated.

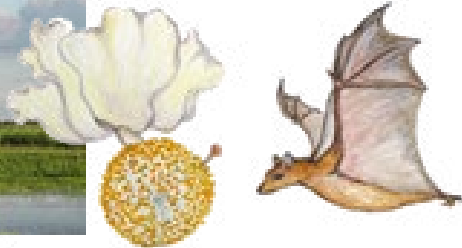


Oxpecker and giraffe

The oxpecker gets food by picking food (ticks and lice) off the giraffe. The giraffe stays disease-free.



Bats live in baobab trees and eat its fruit. In turn, the seeds of the baobab fruit are dispersed.



Bats and baobabs



A special, mutually beneficial, relationship exists between a small bird, the honey guide, and the honey badger. Both derive food from bee nests. The bird has no trouble finding these nests, but it cannot get inside them.

The honey badger, on the other hand, has difficulty finding the nests, but with its strong claws and muscular jaws, the honey badger can easily rip the hives open to get at the honey.

So the bird and the badger join forces. When the bird discovers a wild bees' nest, it searches out a badger and chatters loudly and persistently. The badger answers by moving towards the bird, replying with chuckling and hissing sounds.

The bird leads the way to the hive. On arrival at the hive, the badger breaks into it and begins to feast on the honey and larvae. The bird waits patiently for its turn at the left overs and beeswax.



Honey guide (bird) and honey badger

Commensalism

What is commensalism?



Oysters and mangrove tree roots

Oysters are often found attached to the aerial roots of mangrove trees. Thereby, the oyster has a place to live and grow, and the mangrove tree is not harmed.



Cattle egrets and cattle

Cattle egrets accompany cattle while they forage. As the cattle feed, they disturb the grass and expose insects and small vertebrates. The egrets therefore feed more easily when they accompany the cattle. The cattle neither gain nor are they harmed.



Sharks and remora fish

Remora fish attach themselves to sharks. This means that the fish uses up less energy to move around and gains protection. It also catches the food that the shark drops. The shark does not gain or lose.



Birds and trees

Birds gain a safe nesting place in a tree, and the tree does not gain and is not harmed.

Parasitism

What is parasitism?

Internal parasites are called endoparasites. They live inside the body of the host.

External parasites are called ectoparasites. They live outside the host.

The host is harmed but usually not killed. Why?



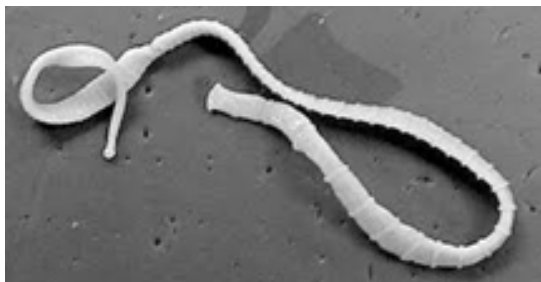
Ticks – ectoparasites

Ticks attach themselves to a host, such as cattle or humans, and feed on the blood. Ticks sometimes carry bacterial infections and transmit the bacterial infection to the host, e.g. tick bite fever. The tick benefits by getting food and the host is harmed.



Fleas – ectoparasites

Dogs and cats are prone to flea infestations. Fleas use their host's blood as food. If untreated, the host will become anaemic due to blood loss and develop skin problems. The flea benefits by getting food and the host is harmed.

**Tapeworms - endoparasites**

Tapeworms are flattened worms that live in the small intestine of the host, such as a human. The eggs of a tapeworm enter the host through raw or undercooked pork.

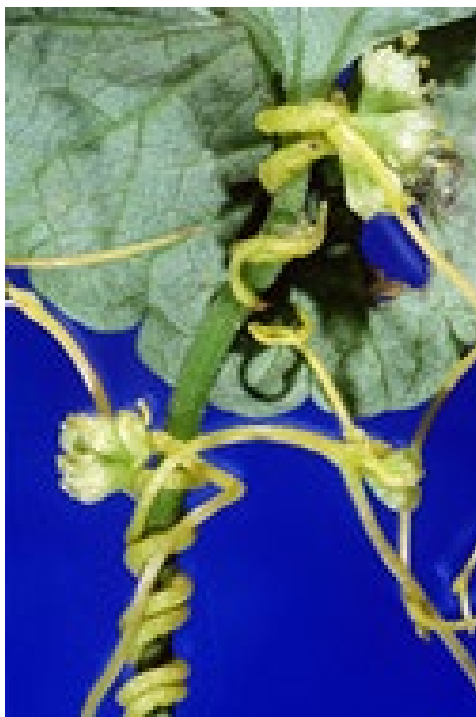
People who have a tapeworm initially experience no symptoms. Later, they experience tiredness, abdominal pain, weight loss and diarrhoea.

The tapeworm benefits by getting food that the host ingests, and the host is harmed.

**Malaria – endoparasites**

Malaria is a parasitic microorganism that is carried by a certain type of mosquito. When a person is bitten by a malaria-carrying mosquito, the malaria parasite enters the bloodstream. Malaria needs to live in red blood cells and it damages them in the process.

The malaria parasite benefits by finding a place to live, and the host is harmed.

**Dodder – plant ectoparasites**

A dodder plant looks like pale yellow spaghetti. A dodder plant seed germinates and immediately looks for a host plant.

There are many different types of host plants for a dodder plant, such as several vegetable plants.

The dodder plant has root-like branches that penetrate the stem of the host plant. This is how the dodder plant gets food and water.

The dodder plant benefits by getting nourishment from the host plant, and the host is harmed because its growth slows down.

CONSERVATION OF THE ECOSYSTEM

Many ecosystems are destroyed due to human activities. We have already discussed above “human factors that disrupt ecosystem balance”. We are now going to discuss why it is important for us to conserve our species.

Why conserve species?

What does conservation mean?

Natural ecosystems carry out many important processes. For example:

- Regulating temperature and climate: This occurs as a result of gas emission from plants of oxygen, carbon dioxide and water vapour from the processes of photosynthesis and respiration.
- Filtering and cleaning water: The vegetation prevents water from running off into rivers. Instead, the water goes into the soil and gets purified by microorganisms.
- Improving the quality of soil: Dead organic matter contributes to soil fertility and quality.
- Recycling waste: Decomposers break down dead plant and animal material and return nutrients to the soil.
- Producing wood: Wood is used for building material and fuel (burning firewood).
- Providing a home: The ecosystems offer a home to all the species that we depend on for pollination, food and medicine.
- Providing high biodiversity: The more natural ecosystems we have, the higher the biodiversity. Having high biodiversity means that if one food source becomes unavailable, we can use others in its place. This has been especially important to people in some rural areas and developing countries. Say there is a disease that affects rice in a rice field in China. If there is a high biodiversity, only some of the rice paddies will be impacted by the disease, not others, because a high biodiversity means that the different rice fields are genetically different, and some are resistant to the disease.



Sustainable use of resources

What does sustainable use of resources mean?

For example, when collecting wood for firewood or fish from a lake for food, only a certain amount should be taken. There should be enough of the resource remaining to allow the population to keep on growing.



Rural villagers catching fish



Rural villagers cutting down wood for firewood

How environmentalists conserve ecosystems

What is the best way to conserve ecosystems?

This includes parks, reserves and marine protected areas such as the Kruger National Park, the Karoo National Park, the Addo Elephant National Park and the Table Mountain National Park. These places are managed in an ecologically sound way. By law, no one is allowed to collect wild plants or remove wild animals, no one is allowed to build on the land and no one is allowed to clear the land.

Alien species have to be removed from ecosystems, because they create an imbalance of the ecosystem and eradicate natural species. Controlling alien animals and plant species is very difficult once they are well established in the ecosystem, but there are three methods that may be introduced:

Biological control – Living things are used to control the population of the alien species. Ecologists look for natural enemies of the alien species in its native environment.

For example, red sesbania is a shrub that produces reddish orange flowers. This species is native to Brazil, Argentina, Paraguay and Uruguay. It has spread to parts of Africa, other parts of South America, and many coastal areas in the southern USA.



Red sesbania

The South American weevil was introduced to South Africa to control the red sesbania. This weevil eats the leaves and flowers of the red sesbania. Thereby, fewer seeds can be formed. The weevil has not prevented new red sesbania from growing, but has radically slowed down their rate of spread.



Considerable research is needed before a biological control species may be let into the environment. What is the biggest worry to ecologists when they introduce a biological control species?

South American weevil eating a leaf

The biological control species needs to be entirely dependent on the alien species that it is meant to eliminate.



Chemical control

Chemical control – This is used only for alien plants (not animals). Chemical poisons are applied to the alien plants. The chemical needs to be applied in such a way that it does not spread and affect other plants. However, the poisons will eventually enter the ecosystem, either when it rains or when the plant dies and it gets decomposed. This needs to be repeated every year because seeds may continue to germinate for many years.

Mechanical control – This involves the killing of alien plants or animals. Plants may be burnt, cut down or uprooted. This needs to be repeated every year because seeds may continue to germinate for many years.



Mechanical control

Sometimes, a combination of these methods is used.

How can I contribute to conservation?

The most practical and simple thing that you can do to help to conserve our natural environment is to recycle.

_____ are recyclable.

Also, never place batteries or electronic equipment in the bin. Take them to a special disposal station.

Recycling means that people produce less waste. Why is this a good thing?

In turn, there will be less damage to the environment.

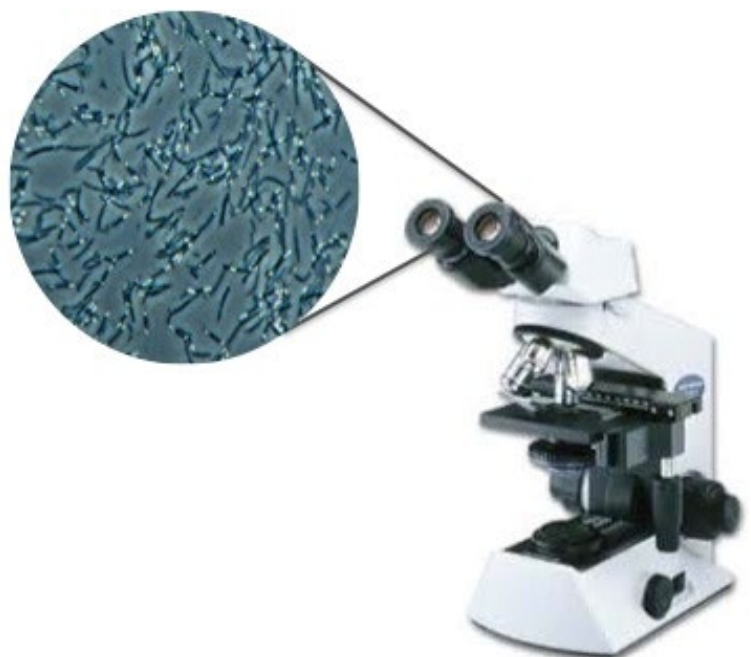


Landfill site

MICROORGANISMS

Introduction

What are microorganisms?

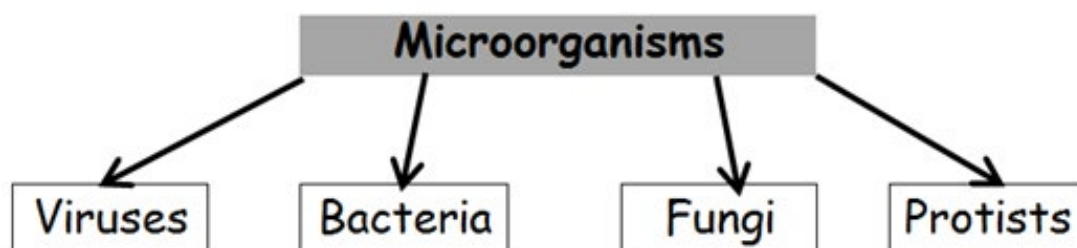


These microorganisms are all around us, but we are unaware of them. We only become aware of them when they make us sick or when our food decays. Before microscopes were invented, people had no idea that microorganisms existed.

Types of microorganisms

There are many different types of microorganisms. They include bacteria, fungi and protists. Viruses are also included when we talk about microorganisms, but viruses are not considered to be living things because they do not display all seven characteristics of living things.

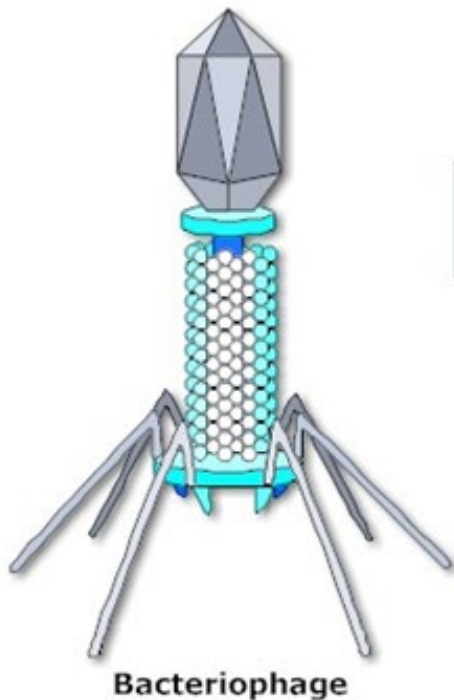
Most microorganisms are unicellular, meaning that they are made of one cell. (Organisms that are made of many cells are multicellular.)



Viruses

Viruses are microscopic particles that attack the healthy cells of living organisms. They are not categorised into any of the five kingdoms of living organisms. This is because they are not able to reproduce on their own. They need to use material from the healthy cell that they have infected in order to reproduce. In this process, they destroy the host cell.

Here are some examples of viruses and their shapes:



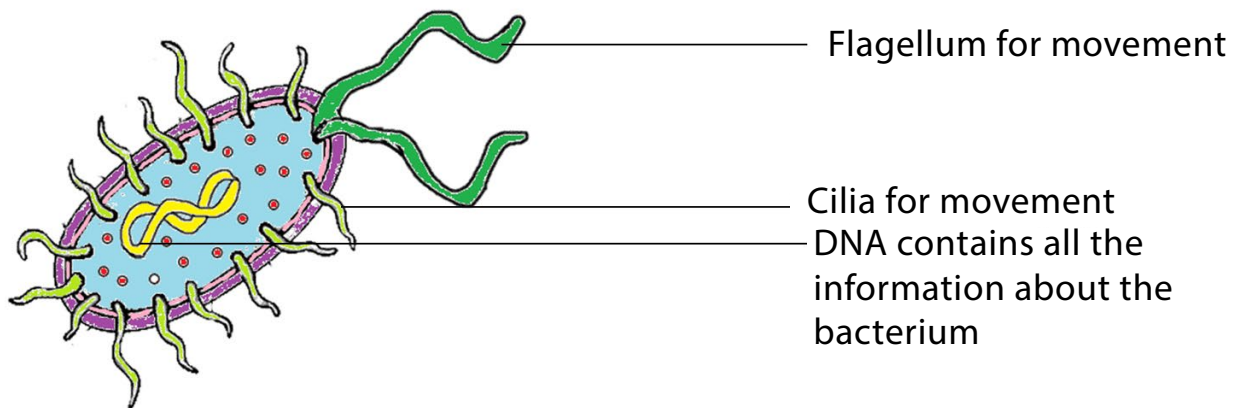
Bacteria

Bacteria are living organisms that are unicellular. There are millions of species of bacteria, but the basic bacterial shapes are rods, spheres or spirals.



Basic bacterial shapes. From left to right: rod, sphere, spiral

Below is a diagram of a simple bacterial cell that is rod-shaped.



Bacteria may be bad for us or good for us. For example, there are bacteria in our mouth that cause cavities. We may get a bacterial infection in our bodies such as a throat or ear infection. But there are also good bacteria. Good bacteria in our bodies help digest our food. Good bacteria are used in making some of the dairy products we like to eat, such as yoghurt, and also some types of medicines. Bacteria are also responsible for decomposing dead organic matter and returning nutrients to the soil.

Protists

Protists are living organisms that may be unicellular or multicellular, and their cell has a more complex structure than that of bacteria. Some of them display more animal-like characteristics and some of them display more plant-like characteristics. Organisms in the protist kingdom include amoebae, red algae, seaweed, green algae, dinoflagellates and diatoms.



Amoeba



Red algae



Seaweed



Green algae

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Dinoflagellate



Diatoms

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Fungi

Fungi are living organisms that are mostly multicellular (with the exception of yeast, which is unicellular) and cannot make their own food. They digest food outside of their body. Fungi may be microscopic or macroscopic.

The largest living organism ever found is this honey mushroom, found in a forest in Oregon in the western part of the USA. It covers about 10 km² in its underground structure.



The largest living organism

Most fungi grow on a food source and spread branching filaments called hyphae. These filaments excrete an enzyme that digests the food. The filaments then absorb the digested food.



Hyphae of fungi

Preventing infection with microorganisms

Disease-causing microorganisms are found everywhere and it is impossible to prevent exposure. Things you touch, eat and drink are filled with microorganisms. There are, however, some steps that may be taken to reduce the risk of being infected:

- Wash your hands with soap before eating.
- Cover your mouth when you cough or sneeze.
- Keep food in the fridge to prevent rapid growth of microorganisms.
- Wash fruit and vegetables before eating.
- Do not share personal items such as toothbrushes, drinking glasses or dining utensils.

Remember that we have an immune system that fights off diseases. The more diseases it fights off, the stronger it gets. It is also important to be exposed to harmful microorganisms in order to strengthen the immune system.

Curing disease

In the early 1900s, the average life expectancy was 47 years. Today, the average life expectancy is 78 years. This is due to the development of vaccines and antibiotics. Scientists such as Louis Pasteur and Alexander Fleming made discoveries that led to new ways of preventing infectious diseases.

Pasteurisation

Louis Pasteur was born in France in 1822. In 1864, Pasteur conducted research that proved that microorganisms in liquids such as milk and wine caused them to spoil, making people ill when they drank them. He invented a process that heats liquid so that most of the microorganisms are killed. This process is known as pasteurisation.



Louis Pasteur (1822–1895)

Vaccines

When viruses or bacteria enter your body, your immune system generates antibodies to try to fight it off (a special antibody is made against each harmful microorganism). Some of the antibodies that are created will remain in your body playing watchdog after you are no longer sick. If you are exposed to the same virus or bacterium in the future, the antibodies will recognise it and fight it off.

A vaccine is _____. This is injected into the person and so the person's body make antibodies against this microorganism. In the case of a real infection, the antibodies are there, ready to fight.

Vaccines can be made against viruses such as measles, mumps and chickenpox. Vaccines can also be made against bacteria such as tetanus and tuberculosis.

Vaccines are not 100% effective. People's bodies respond differently to the vaccines and in some cases not enough antibodies are generated to prevent reinfection. For example, the chickenpox vaccine is only 85% effective, but the measles vaccine is 99,7% effective.

Antibiotics

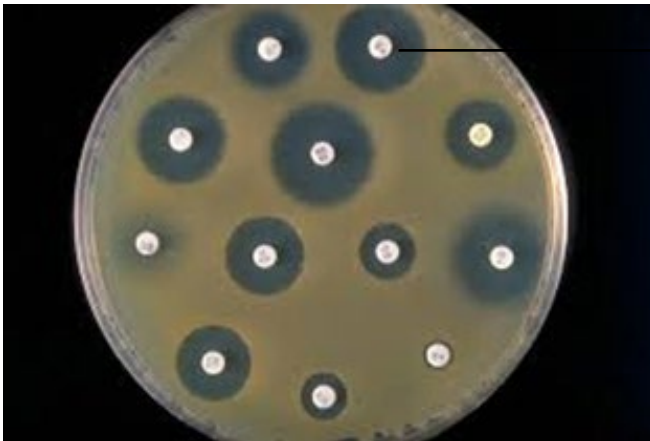
Antibiotics are medicines (such as penicillin) that _____.

Alexander Fleming was born in Scotland in 1881.

In 1928, he was studying staphylococci bacteria. He noticed that mould had



developed accidentally on a set of culture dishes being used to grow the staphylococci germ. The mould had created a bacteria-free circle around itself.



Bacteria-free circle

Bacteria-free circles caused as a result of the mould that accidentally grew on the dishes

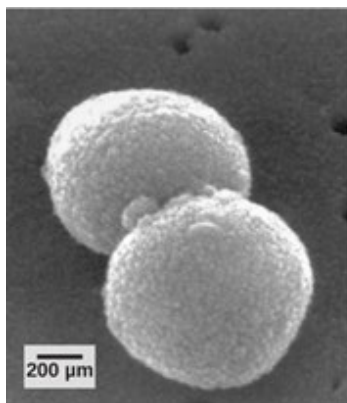
Fleming experimented further and named the active substance penicillin. Penicillin is an acid produced by the mould during its metabolic processes (breaking down food molecules). Many incurable diseases at the time could now be cured with penicillin because penicillin inhibits the growth of many harmful bacteria. Penicillin prevents the cell wall of the bad bacteria from forming, so the bacteria die.

Antibiotics do not work against viruses, only bacteria!

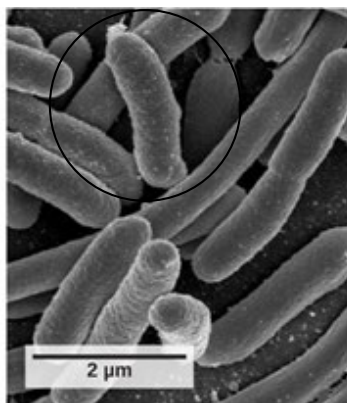
Micrographs and scales

We already learnt that microorganisms are not visible to the naked eye. We use a microscope or a micrograph to see them. A micrograph is a photograph of an object viewed under a microscope. By just looking at a micrograph, we cannot estimate the real size of the organism. Usually, a micrograph has a scale bar. This scale bar allows us to calculate the actual size of the organism in the micrograph.

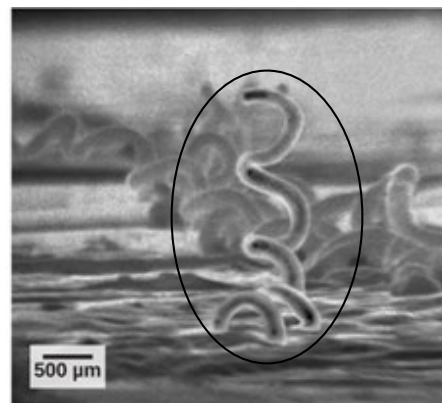
Look at the micrographs below:



(a)



(b)



(c)

Micrograph of (a) sphere-shaped bacteria, (b) rod-shaped bacteria and (c) spiral-shaped bacteria

Consider micrograph (a). You can see the scale bar on each micrograph. This means that the length of the bar is actually equal to 200 μm in real life.

Micrographs will display a bar that's in μm or nm.

- 1 μm = 0,001 mm
- 1 nm = 0,00001 mm

If the bar itself is 6 mm, then anything that is 6 mm long in the micrograph is 200 μm in real life.

To calculate the real length of the organism from a micrograph, follow the steps below:

- Step 1: Measure the length of the scale bar in mm.
- Step 2: Measure the length of the microorganism in mm.
- Step 3: Divide the length of the organism by the length of the scale bar, then multiply your answer by the length printed on the scale bar. Add the unit of the scale bar to your answer.

Let's do this for micrograph (a):

- Step 1: Measure the length of the scale bar in mm: 6 mm
- Step 2: Measure the length of the microorganism in mm. We will use the diameter of the organism in this example: 27 mm
- Step 3: Divide the length of the organism by the length of the scale bar, then multiply your answer by the length printed on the scale bar. Add the unit of the scale bar to your answer.

$$\begin{aligned}
 x &= 27 \text{ mm} \times 200 \mu\text{m} \\
 &\quad 6 \text{ mm} \\
 &= 900 \mu\text{m}
 \end{aligned}$$

Homework 1

Complete this exercise to calculate the actual size of the organism in micrograph (b) and (c). For micrograph (b), use the length of the circled bacterium. For micrograph (c), use the height of the circled spiral bacterium.